

Unmasking Deception: A Novel Approach to Face Swap Deepfake Identification through Geometric Interpolation

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ABSTRACT Deepfake technology has in today's world advanced to the point where face swap manipulations can produce about so indistinguishable forgeries raising serious concerns around privacy, security as well as digital trust. Most existing detection approaches make up the problem as supervised classification making their training on large collections of labeled fake data and although these methods perform well within their training distribution these tend to break down when faced with novel manipulation pipelines or any cross-dataset evaluation. We take a fundamentally different approach. Rather than learning what fakes look like, our framework asks whether a face in a video genuinely belongs to its claimed identity or not reformulating face swap detection as a geometric verification problem in face recognition embedding space. Building on the FACTOR fact-checking paradigm, we have introduced three novel scoring methods that require no training on deepfake data altogether. Our primary contribution Inter-Identity Betweenness Geometry models the face swap deepfakes as geometric interpolations between source and target identity centroids in embedding space, thus achieving 99.99% AUC on Celeb-DF v2 and 99.97% on FaceForensics++ Deepfakes. We additionally propose Negative Margin Scoring, which detects source identity leakage by comparing similarity to the claimed identity against competing identities claiming 99.90% AUC on Celeb-DF v2, and Statistical Identity Modeling, which replaces the single maximum similarity used by FACTOR with the full distribution of reference similarities (99.80% AUC). Evaluation spans two standard benchmarks comprising 11,935 videos across 1,059 identities, four manipulation types, and two compression levels. This method of using betweenness is very astonishingly efficient as it needs as few as four reference frames per identity verification while dropping only 0.03% AUC under heavy compression media. Source identity detection analysis confirms that our geometric framework correctly identifies the actual donor face used to create a deepfake in 96.2% of cases given, providing direct validation of the underlying theory. We provide comprehensive ablation studies, bootstrap confidence intervals, failure case analysis, and per-manipulation breakdowns. In the interest of reporting, we noted that identity based methods fundamentally cannot detect reenactment manipulations such as Face2Face and NeuralTextures, since these preserve the subject's identity rather than replacing it.

INDEX TERMS Deepfake detection, face-swap detection, identity verification, face recognition embeddings, geometric embedding analysis, compression robustness

I. INTRODUCTION

DEEPFAKE generation technology is powered mostly by generative adversarial networks (GANs) [1] and autoencoder architectures [2] that have reached a level of sophistication where face swap manipulations can produce near almost indistinguishable forgeries. What started as a small scale research ,curiosity has grown it into a genuine societal concern [3], [4]. Fabricated videos of political figures along with multiple other cases have appeared on social media platforms, designed to change public opinion during elections [5] [6] . Even some cases of non consensual face swapped videos have caused severe as well as irreversible personal harm to victims [7]. This large scale of the problem is staggering even by some estimates over 95% of deepfake content online is non consensual content, and the total volume of manipulated media doubles over every six months [8]. These numbers paint a troublesome picture, one where the tools of these manipulation are advancing faster than the tools of detection unlike past decades. Even with all downsides it would be intellectually wrong to frame deepfake technology as purely malicious in today's world. The very same generative architectures is what that enable face swapping in legitimate and beneficial applications. Film studios very frequently use face synthesis for de aging actors or completing shoot after an actor's death. Educational content creators online leverage this face animation to bring historical figures to life in ways that textbooks never could to make studying easier. Virtual reality as well as telepresence applications depend on realistic facial rendering for immersive experiences given to user [8], [9]. The challenge, therefore is not just to suppress the technology itself, which would rather be both impractical and counterproductive ,the aim is to rather to develop a reliable methods that can distinguish authentic media from manipulated content when the distinction has a greater value.

The research community has responded with considerable energy regarding this issue [10]. Over the past several years, a very rich and diverse ecosystem of detection methods have emerged and evolved, spanning from convolutional neural network (CNN) based binary classifiers [11], [12] to recurrent architectures that are capable of capturing temporal inconsistencies across video frames [13]. One highly praised method is frequency domain approaches that exploit spectral artifacts which are introduced during synthesis of deepfake content [14] and more recently came the vision transformer architectures that model long range spatial dependencies [15]. Large scale community efforts have greatly accelerated this progress. The Deepfake Detection Challenge (DFDC), organized by Facebook AI in partnership with Microsoft and several other academic institutions, produced one of the largest publicly available deepfake benchmarks with over 100,000 manipulated videos [16]. The FaceForensics++ dataset [12], containing over 1,000 original videos manipulated by four distinct techniques and has become perhaps one of the most widely used evaluation standard in the field.

The Celeb DF dataset [17] raised the bar further by generating very high quality celebrity face swaps with largely reduced visual artifacts as compared to earlier benchmarks of decade. Still despite these impressive body of work, there is a very persistent and somewhat frustrating limitation that still remains. The overwhelming amount of majority existing detection methods are supervised classifiers i.e trained on carefully curated collections of real and fake data that why they tend to degrade sometimes dramatically when confronted with manipulation techniques that absent from their training distribution data [18]. A FaceForensics++ based trained detector has high chances of giving out low scores on Celeb-DF dataset forgeries and there is simple analogy behind it that the pipeline of synthesis is different in both , the classifier will not be able to learn patterns of latter. A major performance drop was observed by Rössler et al. [12] when testing across different manipulation types even within their own dataset. A widely considered and strong supervised baseline the XceptionNet model achieves around 99% accuracy on seen manipulation media but tends to fall below 80% on unseen ones [12]. Data augmentation, self blending strategies [19] or even meta-learning approaches [20] were also used by some researchers to address this concern but the fundamental issue of a classifier being only generalized as its training data still persists.

The issue of generalization have given up a way for new class of thinking overall and this has changed the question from "does this video contain artifacts consistent with known fakes?" to "does the face in this video genuinely belong to its claimed identity?". What the fake looks like is what the first question points to , which changes with every different type of synthesis technique. The latter question requires only the knowledge about what a particular person looks like in real and that remains stable regardless of what or how deepfake technology evolves. Reiss et al. [21] proposed FACTOR framework and worked on this insight by computing cosine similarity between test frame embeddings and that of a trusted reference dataset of the claimed individual and thus achieved competitive detection performance without ever training on deepfake data. FACTOR method relied on identity consistency rather than artifact recognition making it immune to the specific manipulation technique used and at least for face swap attacks where the subject's identity is replaced with another. Despite all this there are gaps in this which our work addresses mainly the scoring mechanism which has a very notable limitation that becomes apparent upon closer examination. The framework relies solely on the maximum cosine similarity between a test embedding and the reference set given all in all asking only a single question "does this face look like the claimed person?". What that model does never considers is whether the face simultaneously resembles some other identity in the database. This is what precisely the geometric signature that face-swapping produces. When a face swap algorithm blends a source (donor) face onto a target identity face the resulting embedding does not simply move away from the

target centroid in some arbitrary direction. Instead, now here is interesting part it moves toward the source identity, creating a predictable geometric pattern in embedding space that can be very useful for detection purpose. The test embedding comes to rest exactly on or near the line connecting the two identity centroids. This is an interpolation artifact that max similarity scoring cannot fully capture all alone. Consider the analogy of a color wheel. Asking "is this color red?" might give assumption a confident yes for both pure red and for a reddish orange that is halfway between red and orange. But on asking "where does this color sit between red and orange?" that immediately gives the idea and reveals about the blend. Our betweenness geometry operates on exactly this principle but in 512 dimensional face embedding space rather than a color wheel.

A. SCOPE AND PROBLEM DEFINITION

Before we go into details it is important to define clearly what our work claims and what it does not. Primarily the issue we address is *identity-referenced face-swap detection* i.e. given a content that says to be a particular individual along with the trusted reference set media of that individual we can find whether the video or image has been manipulated or not using face swapping techniques. This methodology applies to N number of scenarios whether it is in video call authentication, or an organization looking to verify person identity in content or call by checking that he/she is who they claim to be by checking against enrollment footage. In environment or journalism one can check whether a content showing a public figure is authentically sound by comparing it against a dataset of public figures. In today's world of social media moderation platforms can flag suspected face swap media that are going for high profile individuals. In terms of legal consideration forensic testers can validate the true nature of the video proof by verifying the identity consistency against that of independent reference material.

Many of the important boundaries that need to be stated are: First and foremost, the model requires a proper reference dataset of the claimed identity. If there is someone that comes up in a video and there is no reference material for that particular individual our model simply can not make a determination. Apparently this downside is not just limited to our method it comes up with any identity verification technique but it does constraints the deployment scenario to know individual context. Second and seemingly more important thing is methods related to reenactment manipulations such as Face2Face and NeuralTextures [12] that change the person's facial expressions or textures while keeping their identity same are thus undetectable by any identity based approach present out there. As these manipulation tend not to change who the person is only the part of their appearance they go out of scope for detection. We in later part that is Section VI show evidence regarding this limitation where both our presented techniques and FACTOR baseline gave low results of reenactment datasets.

We consider these findings essential for scientific integrity and for any guiding practitioners towards right use cases.

B. MOTIVATION AND RESEARCH GAP

The root of our motivation comes from specific observation that how face swap deepfake exist in embedding space. Recognition models of the modern time like ArcFace [22] are trained of thousands of identities and give a 512-dimensional embeddings where every individual takes a tight cluster of the space. In a face swap algorithm when one replaces identity of A with B, the resulting face tends not to be the exact match of B rather it comes up as blend which is more like structural characteristics of A's face (shape, skin tone, sometimes hair) along with B identity calling features (eye configuration, nose shape, facial proportions). The blend vector does not sit perfectly in B's cluster or even that of A in embedding space instead it comes up in position somewhere between the two. Effect of interpolation is not just theoretical. we verified it using source identity detection that is by checking whether the nearest non claimed identity centroid still corresponds to the actual source identity used to create a given deepfake. During this we found agreement rates of 96.2% on FaceForensics++ Deepfakes and 93.0% on FaceForensics++ FaceSwap dataset. The geometric interpolation is thus very real, measurable and can be called very remarkably consistent across different face swap techniques. The research gap is thus very straightforward. Existing identity based detection methods like FACTOR chief among them and ask only whether a test face matches the claimed identity. They do not put up the question about inter identity geometric structure of the embedding which carries a strong and exploitable signal for face swap detection. To our knowledge no prior model has formulated deepfake detection as geometric interpolation detection in face recognition embedding space. Our betweenness geometry directly addresses this gap for detection purpose.

C. CONTRIBUTIONS

Considering FACTOR as baseline and upon building on it, our model proposes three novel scoring methods that exploit the inter identity geometric structure of a given face recognition embeddings. All methods are fully unsupervised meaning they require zero labeled deepfake data, perform zero gradient updates and involve no classifier training whatsoever. The pretrained used in our model ArcFace ResNet-50 [22], is trained on the WebFace600K dataset that comprises of over 600,000 identities and 10.6 million face images and thus used solely as a frozen feature extractor. Face detection is performed using InsightFace RetinaFace [23]. Our contributions are listed below:

- 1) **Inter Identity Betweenness Geometry:** We introduced a geometric framework that models face swap detection as identity interpolation detection in embedding space itself. For each of the test frame embeddings e , we compute its projection parameter t onto the line connecting the claimed identity centroid

c_{claimed} and the nearest other identity centroid c_{other} . Real faces consistently yield $t \approx 0.84\text{--}0.92$ telling us the proximity to the claimed identity whereas the face swap fakes yield $t \approx 0.38\text{--}0.48$ reflecting their position between two identities. This technique achieves 99.99% AUC on Celeb-DF v2 and 99.97% on FaceForensics++ Deepfakes (c23 compression), with a bootstrap 95% confidence interval of [100.00%, 100.00%] on Celeb-DF v2. In our domain knowledge this model is the first work of deepfake detection as geometric interpolation detection in face recognition embedding space.

- 2) **Negative Margin Scoring:** The FACTOR baseline only compares a test face to the claimed identity's reference embeddings. We additionally check it against the k nearest other identities (with $k=5$ in our experiments). This computes the margin between the maximum similarity to the claimed identity and the maximum similarity to the nearest competing identity. Face swap fakes usually carry residual source identity features that even elevate their similarity to the source thereby reducing the margin. This approach achieves 99.90% AUC on Celeb-DF v2 which is a meaningful improvement over the FACTOR baseline's 99.40%.
- 3) **Statistical Identity Modeling:** Where FACTOR summarizes the similarity between a test frame and the reference set using only the maximum value, we model the full distribution of cosine similarities across all reference embeddings for the claimed identity. Specifically, we compute the mean similarity, which captures overall alignment rather than best-case matching. This seemingly simple change yields 99.80% AUC on Celeb-DF v2, because genuine faces tend to be consistently similar to all reference embeddings, while face-swap fakes show higher variance and lower mean similarity due to imperfect identity transfer.
- 4) **Source Identity Detection (Analysis Contribution):** Giving a direct validation of the geometric interpolation theory underlying our betweenness method we tested whether the nearest non claimed identity centroid for each fake video corresponds to the actual source (donor) identity used during deepfake synthesis. Our finding got the agreement rates of 96.2% on FaceForensics++ Deepfakes, 93.0% on FaceForensics++ FaceSwap, 94.6% on Celeb-DF v2 and 87.3% on FaceForensics++ Deepfakes even under heavy compression (c40). These rates provide significant evidence that our model detects the true manipulation mechanism of identity interpolation rather than just incidental statistical patterns.

A Spearman rank correlation analysis as conducted between our techniques and the FACTOR baseline brings up that betweenness achieves a correlation of only $\rho = 0.729$ with FACTOR confirming that it captures substantially

different information about the test embeddings. The second margin method correlates at $\rho = 0.930$ with betweenness indicating a related but non redundant signal and at end statistical modeling shows $\rho = 0.785$ with betweenness. We validated all these contributions on two standard benchmarks that are Celeb-DF v2 [17] (59 identities, 5,935 videos) and FaceForensics++ [12] (1,000 identities, 5,000 videos spanning into four manipulation types mainly Deepfakes, FaceSwap, Face2Face, and NeuralTextures). We also performed additional compression robustness testing on FaceForensics++ at heavy compression (quantization parameter 40). A comprehensive ablation study of these examines six candidate methods, three of which (temporal consistency scoring, multi-layer feature fusion and adaptive thresholding) were tested and found to be ineffective altogether thus are reported along with detailed analyses of their failure modes and reasons. Reference set size ablation analysis suggest that the betweenness method requires as few as four reference frames per identity while maintaining 99.98% AUC. This is a finding with significant practical implications for deployment. Bootstrap 95% confidence intervals, frame-level versus video-level metric comparisons, computational cost analysis and detailed failure case examination are provided throughout for validation and verification.

II. LITERATURE REVIEW

The deepfake detection landscape has evolved very rapidly over the past half decade or so and is shaped by an arms race between increasingly sophisticated generation techniques and the methods designed to counter them. In this section, we thoroughly survey the generative techniques that are used to produce these deepfakes, the supervised detection paradigms that dominate this field, the emerging identity based alternatives that motivate our own approach and model and finally the benchmark datasets against which all methods are measured and tested. We close by summarizing the specific gap that our contributions address in the domain.

A. DEEPAKE GENERATION TECHNIQUES

Understanding how these deepfakes are created is in many ways kind of a prerequisite for understanding why detecting them is so difficult and why they require heavy computation. The earliest face swap systems relied heavily on autoencoder architectures [2]. This is where a shared encoder learns a common facial representation at the same time a separate decoders reconstruct the faces of two different individuals. By swapping decoders at inference time one can say these systems generate a face that carries the identity of one person like A rendered onto the facial structure of another say as B. The original DeepFakes method which lent its name to the entire phenomenon followed exactly this recipe to synthesise content. Generative adversarial networks also called GANs [1] moved the quality of the ceiling considerably higher from its predecessors. A well known StyleGAN family [24]–[26] demonstrated that any synthesized face could become feasibly and virtually indistinguishable from

real photographs to the untrained eye. FaceSwap one of an open source tool considers taking a more geometric approach by fitting 3D face models and then performing texture mapping. Face2Face and NeuralTextures [12] operate entirely differently from face swap methods. They in a way matter very enormously for detection, talk about modify facial *expressions* or *textures* while preserving the subject's identity they do it all. This distinction between identity replacing manipulations (face swap) and identity preserving manipulations (reenactment) is one we return to throughout this paper, because it defines the core boundary of application for any identity based detection method, our model included. A more recent talk is the diffusion models [27] which have begun producing synthetic faces of remarkable quality and diversity. These models pose new challenges to detectors that are developed primarily against GAN based forgeries. The analogy is that the generation landscape is somewhat a moving target and any detection approach that depends on recognizing generator specific artifacts for detection is at risks of obsolescence with each new synthesis technique that comes out lately [28].

B. SUPERVISED DETECTION METHODS

The dominant technique in deepfake detection remains to be the supervised binary classification. In this type we train a deep network on labeled real and fake data then follow up with its deployment which is further used to classify unseen samples. This approach has produced very impressive in-domain results and thus is considered very effective but at the end it carries a structural vulnerability that becomes more and more visible as the field matures with time. Self-supervised pretraining [29], self-consistency learning [30], high frequency feature analysis [31] or any fine-grained forgery clue mining [32] represent additional recent directions in the world of detection deepfakes. Still cross-dataset Celeb-DF performance on any of these methods generally remains below 90% which marks for improvement.

1) CNN-Based Approaches

Afchar *et al.* [11] proposed MesoNet which is a compact CNN targeting the means for real-time deployment. With its shallow architecture it achieved around 89% accuracy on its training distribution but nonetheless became limited because of its ability to capture the subtle artifacts that newer generators produce. Rössler *et al.* [12] adapted XceptionNet [33] for deepfake detection purpose, achieving about 99% accuracy on FaceForensics++ when trained and tested on the same manipulation type. This number, however, hid a very critical weakness that on Celeb-DF v2 in cross-dataset evaluation that XceptionNet drops roughly to around 65.3%, a decline of 30 plus percentage points [17]. Many other CNN architectures have been also explored including constrained convolutional layers by Bayar and Stamm [34], steganalysis inspired features by Cozzolino *et al.* [35] and VGG16-Siamese combinations by Amerini *et al.* [36]. Each of these model works reasonably within its training distribu-

tion but none of them has convincingly solved cross-dataset generalization problem.

2) Temporal and Sequence Based Methods

Video deepfakes offer a temporal dimension that several researchers have sought to exploit specifically for deepfake detection. Güera and Delp [13] proposed one of the first CNN and LSTM hybrids which extracted spatial features per frame and modeled their temporal evolution. Sabir *et al.* [37] extended this technique with a 3D CNN and bidirectional LSTM reaching up to 85.1% on FaceForensics++. Li *et al.* [38] getting insights after this demonstrated that early deepfakes often exhibited unnatural blinking patterns in them even though this cue has been largely neutralized by modern generators that explicitly model eye behavior for better deepfakes content. The temporal consistency hypothesis is one that deserves particular attention because it gets connected directly to one of our own findings which we used in model. Our ablation experiments with temporal consistency scoring (Section VI) confirm that these modern high quality deepfakes maintain a frame-to-frame coherence which is essentially indistinguishable from real videos. This achieves only 64% AUC on Celeb-DF v2 and dropping to 40% under heavy compression. Temporal smoothness as it appears, is no longer a reliable discriminator. Yang *et al.* [39] similarly exploited head pose inconsistencies in media as a detection signal.

3) Frequency, Attention and Transformer Methods

Qian *et al.* [14] introduced the world of detection with frequency aware detection method. On observing that deepfakes often exhibit unnatural spectral characteristics that are invisible to spatial-only methods. Their approach achieved 91.0% AUC on Celeb-DF which was a meaningful advance though still far from near-perfect detection that is needed in today's scenario. The same group's F3 Net [40] further explored this frequency statistics combining it with local frequency features which dragged it to achieving 97.5% in in-domain on FaceForensics++. Although this was dropped to approximately 65% in cross-dataset Celeb-DF evaluation. Liu *et al.* [41] proposed SPSL which leveraged spatial phase spectrum information reaching to mark of 76.9% on Celeb-DF cross-dataset. Zhao *et al.* [42] proposed a multi-attentional network that concentrates on discriminative local regions such as blending boundaries and texture transitions but even with these sophisticated approaches reached only 67.02% on Celeb-DF in cross-dataset evaluation. Nguyen *et al.* [43] applied capsule networks achieving 84% on FaceForensics++ and roughly 57.5% on Celeb-DF dataset. Vision transformers also referred as ViTs [15] have also been adapted to this task of detection. Wang *et al.* [44] demonstrated that self-attention can model very long range spatial dependencies but gets empirical gains over well-tuned CNNs which have been modest in cross-dataset settings. This is partly because transformers require more training data than what a typical deepfake benchmarks

provide. Xu et al. [45] proposed TALL Swin, which on combining thumbnail level features with a Swin Transformer backbone produced 90.79% on Celeb-DF which is competitive but still well below identity based methods results.

4) Self Blending and Augmentation Strategies

Recognizing that this generalization bottleneck stems partly from the limited forgery diversity during training period several of the groups have explored the idea of creative data augmentation. Shiohara and Yamasaki [19] proposed a Self Blended Images (SBI) that generated pseudo forgeries by blending a face with a transformed version of itself. SBI achieves 93.18% AUC on Celeb-DF without requiring any actual deepfake training data making it one of the strongest supervised cross dataset results up to date. Li et al. [18] introduced Face X-Ray which detected blending boundaries rather than identity specific artifacts which made it achieve 93.47% on Celeb-DF with blending based augmentation. Cao et al. [46] proposed RECCE using reconstruction learning method to capture compact representations of real faces but it achieved only 68.71% in cross dataset Celeb-DF evaluation. These augmentation based methods overall represent the current frontier of supervised generalization. Yet even with the best among them like Face X-Ray at 93.47% it falls meaningfully short of a near perfect performance that is achievable when detection is reformulated entirely around identity verification.

5) The Generalization Challenge in Context

The pattern seen across supervised methods is a consistent one and worth stating plainly. In-domain FaceForensics++ results routinely the methods reach around 95–99%, while with cross dataset Celeb-DF results for the same methods typically fall to 55–75%. With even the most carefully designed approaches rarely exceeding 93%. The problem is not architectural here the transformers, attention mechanisms and frequency analysis are all very capable tools. The problem is that the supervised classifiers learn to distinguish real from fake based on the specific artifacts present in their training data from which they get bound and those artifacts change with every new generator, compression scheme or post-processing step. Data augmentation and self blending mitigate this to some degree but even they cannot fundamentally resolve it. Zero shot approaches [47] and color cue methods [48] have also been explored but none them have closed the cross dataset gap. Table 1 summarizes the detection methods discussed in above section. We prioritize the cross dataset Celeb-DF v2 AUC values where available as these are the one which most transparently reveal generalization capability. The gap between the best supervised cross dataset results (~93%) and identity based approaches (>99%) is not incremental, it reflects a fundamentally different relationship to the generalization problem altogether.

C. IDENTITY BASED DETECTION

A fundamentally different detection philosophy asks not what fakes look like but whether the face in a given video is consistent with its claimed identity. The answer to "does this face match person X?" does not depend on how a hypothetical fake was created but depends only on what person X actually looks like in real life. The baseline we used in our work FACTOR framework by Reiss et al. [21] formalized this insight. Given a test video claiming to depict a specific individual the FACTOR extracts face embeddings using a pretrained face recognition model and then computes the maximum cosine similarity between each test embedding using a set of trusted reference embeddings for the claimed identity. In the process if the similarity falls below a specific threshold the video is flagged as potentially manipulated deepfake. The main appeal lies in what FACTOR does *not* require that is what makes it good no deepfake training data, no gradient updates and even no classifier training of any kind. The face recognition model the ArcFace [22] is used in both their implementation and ours. It serves purely as a frozen feature extractor in pipeline. FACTOR achieves 99.40% AUC on Celeb-DF v2, already surpassing every supervised cross dataset result in the community by a considerable margin specifically on benchmarks like LFW [56]. This performance advantage has a clear very explanation. Supervised methods must generalize across generation techniques which vary unpredictably depending on data used. Identity based methods needs only to generalize across *appearances of the same person*, which vary in well understood ways (lighting, pose, expression, aging) that modern face recognition models can handle with remarkable robustness.

ArcFace that is trained on the WebFace600K dataset spanning over 600,000 identities and 10.6 million images [22] produces a 512 dimensional embeddings where the individuals form a tight, well separated clusters. The additive angular margin loss used during the ArcFace training is to explicitly maximizes the interclass separation and the intra class compactness. These properties are those that turn out to be directly exploitable for deepfake detection even though the model was never trained with that application in mind. ArcFace builds on the facts of angular margin losses including the SphereFace [57] and CosFace [58] which all designed to maximize inter class separation on a hyperspherical manifold. Nirkin et al. [53] explored a related idea, detecting manipulations by finding discrepancies between the face region and its surrounding context. While effective in some settings, this approach still requires supervised training on labeled manipulation data and does not exploit the inter-identity geometric structure of face recognition embeddings. Dong et al. [59] recently explored identity consistency transformers for protecting celebrities from deepfakes, though their approach still requires supervised training. What the FACTOR method leaves out is to examine the relationships *between* the identities in

TABLE 1. Summary of deepfake detection techniques. The Celeb-DF column shows cross-dataset (CD) AUC where available, as it best reveals generalization. FF++ values are in-domain (ID) unless noted. Our betweenness achieves 99.99% on Celeb-DF v2 with zero deepfake training data. Values from original publications; see [49] for standardized comparisons.

No.	Method	Year	Core Technique	FF++ (%)	Celeb-DF (%)	Training
<i>CNN-Based Classifiers</i>						
1	MesoNet [11]	2018	Meso-4 lightweight CNN	89.0 ID	54.8 CD	Supervised
2	XceptionNet [12]	2019	Depthwise separable conv.	99.0 ID	65.3 CD	Supervised
3	Constrained CNN [34]	2016	Manipulation-aware conv. layer	82.9 ID		Supervised
4	Steganalysis CNN [35]	2017	Residual-based descriptors	78.5 ID		Supervised
5	VGG16 + Siamese [36]	2019	Feature comparison network	81.6 ID		Supervised
6	FakeSpotter [50]	2020	Neuron coverage monitoring		66.8 CD	Supervised
<i>Temporal and Sequence-Based</i>						
7	CNN + LSTM [13]	2018	Spatial-temporal hybrid			Supervised
8	Eye-blink LSTM [38]	2018	Physiological cue detection	89.0 ID		Supervised
9	3D CNN + Bi-LSTM [37]	2019	Recurrent convolutional	85.1 ID		Supervised
10	CNN-LSTM hybrid [51]	2020	Temporal feature fusion		77.6 CD	Supervised
11	CNN-LSTM transfer [52]	2020	Cross-dataset transfer		84.0 CD	Supervised
<i>Frequency, Attention, and Transformer</i>						
12	Thinking in Freq. [14]	2020	Frequency-aware features		91.0 CD	Supervised
13	F3-Net [40]	2020	Freq. statistics + local freq.	97.5 ID	65.2 CD	Supervised
14	Multi-Attentional [42]	2021	Local attention regions		67.02 CD	Supervised
15	SPSL [41]	2021	Spatial phase spectrum	96.4 ID	76.9 CD	Supervised
16	Capsule-Forensics [43]	2019	Capsule networks	84.0 ID	57.5 CD	Supervised
17	ViT baseline [44]	2023	Vision Transformer	90.2 ID		Supervised
18	TALL-Swin [45]	2023	Thumbnail + Swin Transformer		90.79 CD	Supervised
<i>Augmentation and Blending-Based</i>						
19	SBI [19]	2022	Self-blended pseudo-forgeries		93.18 CD	Self-gen.
20	Face X-Ray [18]	2020	Blending boundary detection		93.47 CD	Self-gen.
21	RECCE [46]	2022	Reconstruction-classification		68.71 CD	Supervised
<i>Other Notable Approaches</i>						
22	Face-Context [53]	2021	Face vs. context discrepancy			Supervised
23	Lip-Sync [54]	2021	Audio-visual synchronization	88.6 ID		Supervised
24	EfficientNet-B3 [55]	2023	EfficientNet + Attention LSTM		85.0 CD	Supervised
25	Meta-learning [20]	2024	Deep feature stacking	94.0 ID		Supervised
<i>Identity-Based (Unsupervised, No Deepfake Training Data)</i>						
26	FACTOR [21]	2023	Max cosine similarity to ref.	99.88 ID	99.40	None
27	Ours (Betweenness)	2026	Inter-identity geometry	99.97 ID	99.99	None

ID = in-domain (trained and tested on same manipulation). CD = cross-dataset (trained on FF++, tested on Celeb-DF). Self-gen. = trained on self-generated pseudo-fakes only. Empty cells indicate the metric was not reported in the original publication.

the embedding space. It asks only a one sided question that whether "does this face match the claimed identity?". It never considers whether the face also simultaneously resembles another identity or not which is precisely the geometric signature that face swapping produces. This is the gap our work addresses in our work.

D. BENCHMARK DATASETS

Reliable evaluation generally depends on a well constructed benchmark datasets and this field has seen a steady progression from small, artifact-heavy collections to large-scale, high-quality and increasingly realistic benchmarks datasets. Table 2 summarizes these datasets which are also used in our evaluation, with particular attention given to what

each dataset or sub dataset is designed to test and why it matters for validating different aspects of a detection systems. FaceForensics++ [12] first introduced in 2019 still remains arguably the most widely used evaluation framework dataset. It contains of about 1,000 original YouTube videos with each of them being manipulated by four distinct types of techniques. Critically these four techniques tend to fall into two categories with very different implications for identity based detection systems. Deepfakes and FaceSwap both perform identity replacement that is the resulting face belongs to a different person and the identity signal in the embedding space is thus disrupted. Face2Face and NeuralTextures use the reenactment technique which is that they alter the expressions or textures while preserving

the subject's identity itself. We evaluate on all four types specifically with empirical evidence to establish that in which cases the identity based methods excel and where they are fundamentally inapplicable. The datasets also provides multiple compression levels (c23, light; c40, heavy) which allows for the testing whether detection survives in the lossy compression environment that real world video distribution inevitably introduces in media. Celeb-DF v2 [17] raised the quality bar substantially in terms of datasets out there. Its synthesis pipeline reduced the common artifacts such as color mismatch, visible boundaries and temporal flickering, making it a considerably harder benchmark than FaceForensics++. With its 59 identities and 5,639 deepfake videos the Celeb-DF dataset has become the standard in industry for evaluating cross dataset generalization in models. When a method trained on FaceForensics++ is tested on Celeb-DF the performance drops are typically dramatic giving often 20-40 percentage points difference. This reveals how much of the detector's original accuracy was tied to dataset specific artifacts rather than just the genuine manipulation understanding of media. The Deepfake Detection Challenge (DFDC) dataset [16] which was produced by Facebook AI in partnership with Microsoft is the largest publicly available benchmark with over 100,000 manipulated videos. FakeAVCeleb [60] introduced multimodal deepfakes including both the audio and video manipulation content with diversity across gender and ethnicity equally. WildDeepfake [61] collected samples from the internet that reflects the real world distribution characteristics. The DeepfakeBench framework by Yan et al. [49] introduced at NeurIPS 2023 gives a unified evaluation platform for standardizing pre processing, training and metrics across 15 detection methods including the 9 datasets. This addresses reproducibility concerns that have complicated the prior comparisons of models.

E. SUMMARY AND POSITIONING

The survey done above reveals a very consistent and notable pattern that the Supervised detection achieves strong in-domain results but that degrades substantially under domain shifts, with even the best augmentation based approaches coming around 93% AUC on Celeb-DF. Identity based detection on the other hand as demonstrated by FACTOR takes some sidesteps for countering the generalization problem entirely by reframing detection problem as identity verification rather than artifact recognition, reaching to about 99.40% AUC without any training. However this FACTOR's scoring mechanism exploits only the similarity between a test face and the claimed identity. It never opens up to examine the inter identity geometric relationships, the negative margin between claimed and competing identities or even the distributional characteristics of reference similarities. Our three contributions address these gaps in depth. In the following section, we describe the proposed methodology in detail.

III. PROPOSED METHODOLOGY

This section describes what our complete detection framework is and how it goes on. We begin with the end to end pipeline that all methods share and then present the FACTOR baseline and its limitations. This is then followed by our three novel scoring methods and the source identity detection analysis that validates about the underlying geometric theory we used in our model.

A. OVERVIEW AND DESIGN PHILOSOPHY

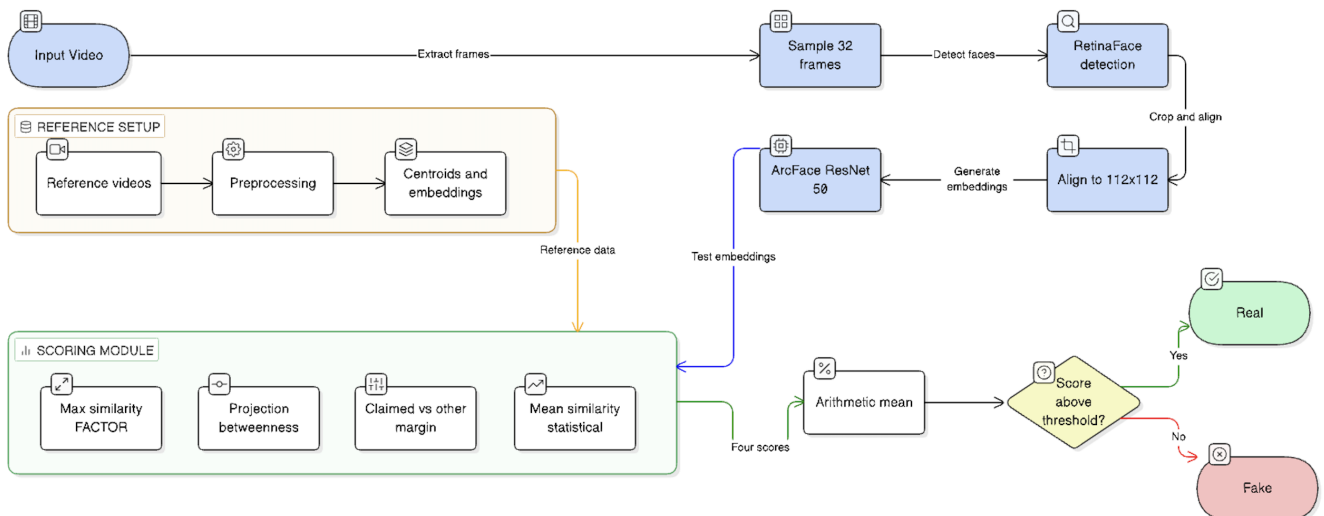
Our approach rests on a simple reasoning that when a face recognition model that was trained to tell people apart in principle should be able to tell when a face has been replaced with someone else's. The entire detection system uses a single pretrained ArcFace ResNet-50 model [22] as a frozen feature extractor for the pipeline. There are no trainable parameters anywhere in our pipeline that need adjustment as well as no deepfake data is required for training of model. No gradient updates are performed here and no classifier is learned. The ArcFace model is originally trained on the WebFace600K dataset comprising over 600,000 identities and 10.6 million face images and was designed for face verification. We repurposed it without modification for deepfake detection. This design by choice is deliberate as it has both philosophical and practical motivations. Philosophically, what we want is the detection system to be agnostic to how deepfakes are generated in modern era. A supervised classifier learns generation specific artifacts which means it can only detect what it has seen or what closely resembles what it has seen in training. Our system by contrast with it only needs to know what a particular real person looks like. If the face in a test video does not match that person's identity profile in embedding space it catches that something is wrong, regardless of what generation technique was used to develop it. Practically, this zero training requirement means our method can be deployed immediately for any individual with even a small collection of reference images and without the computational and data infrastructure that supervised training demands. The complete pipeline used is illustrated in Fig. 1.

B. DETECTION PIPELINE

The pipeline implemented consists of five stages that are shared identically across all scoring methods. Both the test video and the reference material pass through this same pre processing chain ensuring that any systematic biases in face detection or even the alignment affects reference and test embeddings in equally proportion. **Frame sampling from media:** Given an input video first our model uniformly samples out 32 frames from its duration. Uniform sampling ensures that there is temporal coverage without over representing of any particular segment of media. The choice of 32 frames balances between the computational cost against signal reliability in pipeline of method. The ablation analysis as shown in Section VI confirms that this aggregation over

TABLE 2. Benchmark datasets used in this study. FaceForensics++ sub-datasets are listed individually as they represent distinct manipulation types. The "Purpose" column summarizes what each condition validates.

Dataset	Manipulation	Real	Fake	IDs	Quality	Purpose
Celeb-DF v2 [17]	Face-swap (autoencoder)	590	5,639	59	High	Generalization on high-quality fakes; hardest cross-dataset benchmark
FF++ Deepfakes [12]	Face-swap (autoencoder)	1,000	1,000	1,000	c23	Core face-swap detection; scalability across 1,000 identities
FF++ FaceSwap [12]	Face-swap (3D graphics)	1,000	1,000	1,000	c23	Transfer across different face-swap implementations
FF++ Face2Face [12]	Reenactment (expression)	1,000	1,000	1,000	c23	<i>Negative control</i> : identity preserved; expected failure case
FF++ NeuralTex [12]	Reenactment (rendering)	1,000	1,000	1,000	c23	<i>Negative control</i> : texture-only edit; identity preserved
FF++ DF c40 [12]	Face-swap (autoencoder)	1,000	1,000	1,000	c40	Compression robustness; refs from c23, test from c40
<i>Additional datasets (referenced for context; not in our primary evaluation):</i>						
DFDC [16]	Multiple	23,654	104,500	960	Varied	Largest public benchmark; scale and diversity testing
FakeAVCeleb [60]	Swap + lip-sync	500 total			High	Multimodal (audio + video); demographic diversity
WildDeepfake [61]	Mixed (in-the-wild)	7,314 total			Varied	Internet-sourced; real-world conditions
c23 = light compression (QP 23); c40 = heavy compression (QP 40). IDs = distinct identities. Our evaluation uses all six conditions above the double line.						

**FIGURE 1.** End-to-end detection pipeline. The input video is sampled to 32 frames which go through RetinaFace detection, 112×112 alignment, and ArcFace ResNet-50 embedding extraction. The reference setup (left) processes trusted videos through the same chain to produce per-identity centroids and stored embeddings. Both the test embeddings and reference data feed into the scoring module (bottom left), which computes four scores: FACTOR max-similarity, betweenness projection onto the identity line, claimed vs other margin, and mean statistical similarity. The four scores are aggregated via arithmetic mean and compared against a threshold to output a Real or Fake decision with a confidence score.

32 frames helps in providing stable video level scores. If any video contains less than 32 frames our model try to use all available frames.

Face detection: InsightFace RetinaFace [23] process each of the sampled frame ,here in this model it works as a single stage face detector that can both predicts bounding boxes and find the five major point facial landmarks that are present(two eyes, nose tip and two mouth corners). Reti-

naFace achieves this state of the art detection accuracy even under partial occlusion and moderate pose variation. This becomes important because missed detections can create gaps in our per frame scoring sequence that is discussed further. In our implementation this entire pre processing pipeline uses InsightFace exclusively for both face detection and embedding extraction. This ensures full consistency between the reference and test paths. Frames where no face

is detected are silently skipped by model and in practice this occurs in fewer than 1% of sampled frames across both benchmarks datasets.

Face alignment and cropping: Using the five detected landmarks in last step, each of the face is aligned to a canonical frontal pose and then cropped to 112×112 pixels. This is the standard input resolution for ArcFace in which the alignment step normalizes for head rotation and position, ensuring that these embeddings capture identity rather than pose specific information which is not required. ArcFace embeddings are trained to be robust to moderate pose, lighting and even expression variation. The Celeb-DF dataset naturally includes such variation across its celebrity appearances (interviews, red carpets, talk shows filmed from different angles). While this step provides an implicit evidence of pose robustness, systematic evaluation under controlled pose and occlusion conditions for the remains of an important direction for future work (Section VII).

Embedding extraction: The aligned face crops are then passed through a frozen ArcFace ResNet-50 model that produces 512-dimensional L2-normalized embedding vectors. These embeddings occupy a hyperspherical manifold where cosine similarity between vectors directly corresponds to identity similarity. Individuals form tight clusters in space and along with the angular margin loss used during ArcFace training ensures that there is substantial separation between clusters of different identities. It is worth emphasizing on the note that this model is used purely as it is. There is no fine tuning, no adaptation layers or no domain specific modifications. The features that it produces were designed for face recognition specifically not for deepfake detection yet they turn out to be remarkably effective for the latter task primarily because the process of face swapping disrupts the identity signal that these embeddings are optimized to capture.

Reference database construction: For each claimed identity of the media our model construct a reference set by processing trusted video or images through the identical pipeline described in the above sub section (detection, alignment, embedding). On Celeb-DF v2 this process yields approximately 160 reference embeddings per identity from the real celebrity videos in dataset. On FaceForensics++ we use the first 16 frames of each original video are used as reference (the last 16 are reserved for testing). The per identity centroid that is the arithmetic mean of all reference embeddings for that identity is also precomputed and stored simultaneously as it is required by the betweenness geometry method. Reference construction is a one time thing done offline per identity. Our ablation study (Section VI) demonstrates that as few as four reference frames are suffice for 99.98% AUC with betweenness scoring making it a finding with significant practical implications for deployment.

C. FACTOR BASELINE

The FACTOR framework as proposed by Reiss et al. [21] provides our model a starting point and serves as the

baseline against which all our techniques are compared. For each of the test frame embedding e FACTOR computes the maximum cosine similarity between e and the set of given reference embeddings $\mathcal{R}_{\text{claimed}}$ for the claimed identity by the media:

$$S_{\text{FACTOR}}(e) = \max_{r \in \mathcal{R}_{\text{claimed}}} \text{sim}(e, r)$$

Here $\text{sim}(\cdot, \cdot)$ denotes cosine similarity (formalized as Equation-1 in Section IV). Real faces that genuinely belong to the claimed identity here yield high similarity scores. Face swap fakes on the other hand which belong to a different identity provide a lower scores because their embeddings sit in a different region that of real of the embedding space. FACTOR achieves a very strong performance of 99.40% AUC on Celeb-DF v2 and 99.88% on FaceForensics++ Deepfakes. Still upon closer examination the method asks only a one sided question of "how similar is this face to the claimed identity?", it never really considers whether the face also resembles some *other* identity in the given database. This is a meaningful omission that our model focuses on because face swapping does not move embeddings to a arbitrary locations in the space. It tends to moves them toward the source (donor) identity which creates a predictable geometric pattern that max similarity scoring alone cannot fully exploit as seen in FACTOR method. Consider a very specific scenario where Person A's face is swapped onto Person B's video. We can not call the resulting embedding to be a perfect copy of A. It is a blend that retains some of B's structural features while adopting A's identity defining characteristics. This blend often achieves a moderately high similarity to B's reference set if searched and sometimes even high enough to pass FACTOR's threshold. This is because the residual B features tend to keep the embedding partially aligned with B's cluster. But the same embedding simultaneously shows elevated similarity to A's cluster this is signal that FACTOR never checks. Our three methods are designed to capture exactly this kind of inter identity information for detection. Fig. 2 illustrates FACTOR's limitation alongside all the three methods we propose to address these limitation. The left panel shows how FACTOR's single identity check can miss some of the well crafted fake content, while the right panel previews how betweenness, margin and statistical scoring each capture a different signal of the inter identity geometric structure.

D. INTER IDENTITY BETWEENNESS GEOMETRY

Our model primary contribution reframes deepfake detection as geometric interpolation detection problem. This is the highest performing method across all face swap evaluations present in this paper and to our knowledge. The first work is to model deepfake detection as inter identity geometric analysis in face recognition embedding space. This is the core insight that the face swapping creates a embeddings that lie *between* two identity clusters and this positional

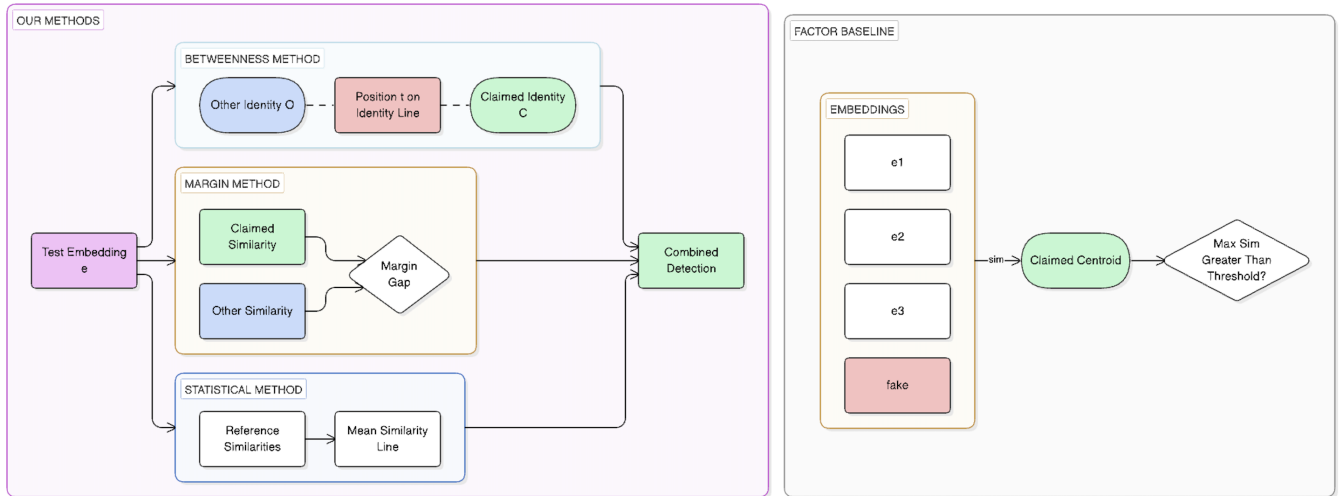


FIGURE 2. Comparison of FACTOR baseline with our three proposed methods. FACTOR (right panel) computes similarity between each test embedding and the claimed centroid only, checking whether the max similarity exceeds a threshold. A well-crafted fake can pass this single check. Our methods (left panel) capture additional inter-identity signals from the same test embedding. The betweenness method measures where the embedding sits on the line between the other identity (O) and claimed identity (C). The margin method compares similarity to the claimed identity against similarity to competing identities and measures the gap between them. The statistical method computes mean similarity across all reference embeddings instead of just the maximum. All three outputs merge into a combined detection score.

information carries a strong and exploitable detection signal for deepfake detection.

1) Geometric Intuition

When a face-swap algorithm replaces identity B (target) with identity A (source), the resulting face is a blend. It inherits A's identity-defining features (eye shape, facial proportions, nose structure) while retaining certain structural characteristics of B (face shape, skin tone, sometimes hair and background facial structure). In the embedding space of the ArcFace, the vector which manifests is the blend that does not sit perfectly in A's cluster nor even in B's cluster. Instead of the both the vector occupies a position somewhere along the direction which connect the two identity centroids. Though it tends to be closer to the source than to the target but not jointly with either. This deduction is not just any theoretical prediction. We validate this through source identity detection as presented in (Section III-G and Section VI-G). About for 96.2% of FaceForensics++ Deepfakes, the nearest non claimed identity centroid relates to the actual donor identity generally which is used to create the deepfake. Thus clarifying about the fact that geometric interpolation is real, measurable, efficiently and are very consistent across different face swap techniques.

2) Betweenness Parameter

To capture the interposition, we defined a betweenness parameter t that allows to captures where a test embedding sits on the line connecting two identity centroids. For each of this test frame embedding e , the first step is find the closest other identity that is the identity in the reference dataset whose centroid c_{other} gives up the highest cosine similarity to e . This excludes any of the claimed identity itself. This

selection is by default performed independently for each of the frame in media, allowing only the method to adapt if any different frames exhibit any different kinds of geometric relationships. We then compute from all this:

$$t = \frac{\text{sim}(e, c_{\text{claimed}})}{\text{sim}(e, c_{\text{claimed}}) + \text{sim}(e, c_{\text{other}})}$$

where c_{claimed} is the centroid of the claimed identity's reference embeddings (formalized as Equation 4 in Section IV). The parameter t ranges from 0 to 1, with $t = 1$ indicating that the embedding is entirely aligned with the claimed identity and $t = 0.5$ indicating a perfect midpoint between two identities. In practice, real faces consistently yield $t \approx 0.84\text{--}0.92$, reflecting strong proximity to the claimed identity with only modest similarity to the nearest other identity. Face-swap fakes yield $t \approx 0.38\text{--}0.48$, reflecting their position between the target and source identity clusters. The gap between these distributions is wide enough to achieve near-perfect separation: 99.99% AUC on Celeb-DF v2 with a bootstrap 95% confidence interval of [100.00%, 100.00%]. The geometric concept is illustrated in Fig. 3.

3) Why This Is Novel

To our knowledge, no prior work has formulated deepfake detection as geometric interpolation detection in face recognition embedding space. Existing identity-based methods, FACTOR included, treat each identity in isolation. Our betweenness geometry explicitly models the *relationship between* identities, which is exactly the structure that face-swapping creates. The color wheel analogy from Section I captures this precisely: asking “is this color red?” misses a blend, but asking “where does this color sit between red and orange?” reveals it immediately.

E. NEGATIVE MARGIN SCORING

Our second contribution extends the FACTOR scoring principle from a one-sided to a two-sided comparison. Where FACTOR asks only how similar a test face is to the claimed identity, negative margin scoring additionally measures how similar it is to competing identities in the database. For each test frame embedding $\mathbf{e}$, we compute the difference between the maximum similarity to the claimed identity's reference set and the maximum similarity to the k nearest other identities ($k = 5$ in all our experiments):

$$S_{\text{margin}}(\mathbf{e}) = \max_{\mathbf{r} \in \mathcal{R}_{\text{claimed}}} \text{sim}(\mathbf{e}, \mathbf{r}) - \max_{j \in \text{top-}k} \max_{\mathbf{r} \in \mathcal{R}_j} \text{sim}(\mathbf{e}, \mathbf{r})$$

Real faces tend to have a big margin that tends to copy more of the claimed identity than the others and small/negatively to the face swap fakes. These swaps go forward to carry a residual of the source identity features that increase their similarity to the identity donor as formalized as Equation 3 in Section IV. The inspiration for this comes very directly. A real face of Person B will be very much equal to B's references and somewhat less similar to anyone else in the reference set. A face swap fake that aims B but is derived from A will lean to show a lessened similarity to B all because the swap is imperfect. The high likeness to A will be due to the reason of source identity features leak through. The margin between these two similarities decreases and this shrinkage is the detection signal for our model. On Celeb-DF v2 the negative margin technique gets to mark of 99.90% AUC which is a solid jump from FACTOR's score of 99.40%. The scores are varied on FaceForensics++ though. The reference set used there is much smaller although only just 16 frames per identity are compared 160 which are used on Celeb-DF. With that size of limited references data margin scoring is set to drop slightly behind FACTOR that is 99.82% versus 99.88%. This does not come surprising really to us. To find out that which other identity is the closest, one need enough reference embeddings to build accurate centroids and even after that 16 frames sometimes is not quite enough for that. What this emphasis practically is that the margin approach works best when you have a decent sized reference database at disposal. For most real world use cases like verifying someone on a video call or checking footage of a public figure having enough reference material is usually not a problem one face.

F. STATISTICAL IDENTITY MODELING

Our third contribution replaces FACTOR's max-similarity aggregation with a distributional approach. Where FACTOR summarizes the relationship between a test embedding and the reference set using only the single highest similarity value, we compute the mean similarity across all reference embeddings for the claimed identity:

$$S_{\text{stat}}(\mathbf{e}) = \frac{1}{|\mathcal{R}_{\text{claimed}}|} \sum_{\mathbf{r}_i \in \mathcal{R}_{\text{claimed}}} \text{sim}(\mathbf{e}, \mathbf{r}_i)$$

This seemingly minor change captures something that maximum similarity misses: consistency (formalized as Equa-

tion 5 in Section IV). A genuine face of the claimed identity tends to be similar to *all* reference embeddings, because natural within-person variation (lighting, expression, slight pose changes) is relatively small in ArcFace embedding space. A face-swap fake might achieve a moderately high maximum similarity to the reference set if one reference embedding happens to be somewhat aligned with the blend, but its average similarity across the full reference set will be lower because the imperfect identity transfer creates systematic misalignment with most references. Statistical identity modeling achieves 99.80% AUC on Celeb-DF v2. Like the margin method, it benefits from larger reference sets: on Celeb-DF (approximately 160 references per identity), the improvement over FACTOR is +0.40 percentage points, while on FaceForensics++ (16 references), it performs comparably to FACTOR. The method is computationally trivial, adding only a mean computation over the reference set, and provides a complementary signal (Spearman $\rho = 0.785$ with betweenness) that could be useful in ensemble combinations.

G. SOURCE IDENTITY DETECTION

Source identity detection is not a scoring method for the real-versus-fake decision. Rather, it is an analysis tool that provides direct empirical validation for the geometric interpolation theory underlying our betweenness approach, and it addresses a natural follow-up question: if we can detect that a face was swapped, can we also identify *whose* face was used as the source? For each fake video, we identify the nearest non-claimed identity centroid: the identity in the database (excluding the claimed identity) whose centroid has the highest cosine similarity to the mean test embedding for that video. We then check whether this identity matches the actual source (donor) identity that was used during deepfake creation. The ground truth source identity is available from the dataset metadata, as both Celeb-DF and FaceForensics++ provide source-target pairings for each fake video. If our geometric interpolation theory is correct, the nearest non-claimed identity should frequently correspond to the actual source, because the face-swap embedding is pulled toward the source identity cluster. This is what our results exactly show. On FaceForensics++ Deepfakes the agreement rate comes out to be about 96.2%. For Celeb-DF v2 it is 94.6%, FaceForensics++ FaceSwap gives somewhat 93.0% and even under heavy compression at c40 it drops to 87.3%. Now these numbers are obviously not 100%. We did not expect them to be, there are cases where the actual source identity is just isn't in the reference database at all. Sometimes two identities look similar enough that they can create confusion. Compression tends to wash out even the finer geometric details that the method tends to relies on. Still, rates remains consistently high and this tells us something important that our betweenness approach is picking up the *real manipulation mechanism* used here that is the identity interpolation itself. This is not just some random statistical pattern that happens to correlate with fakes ones. Beyond

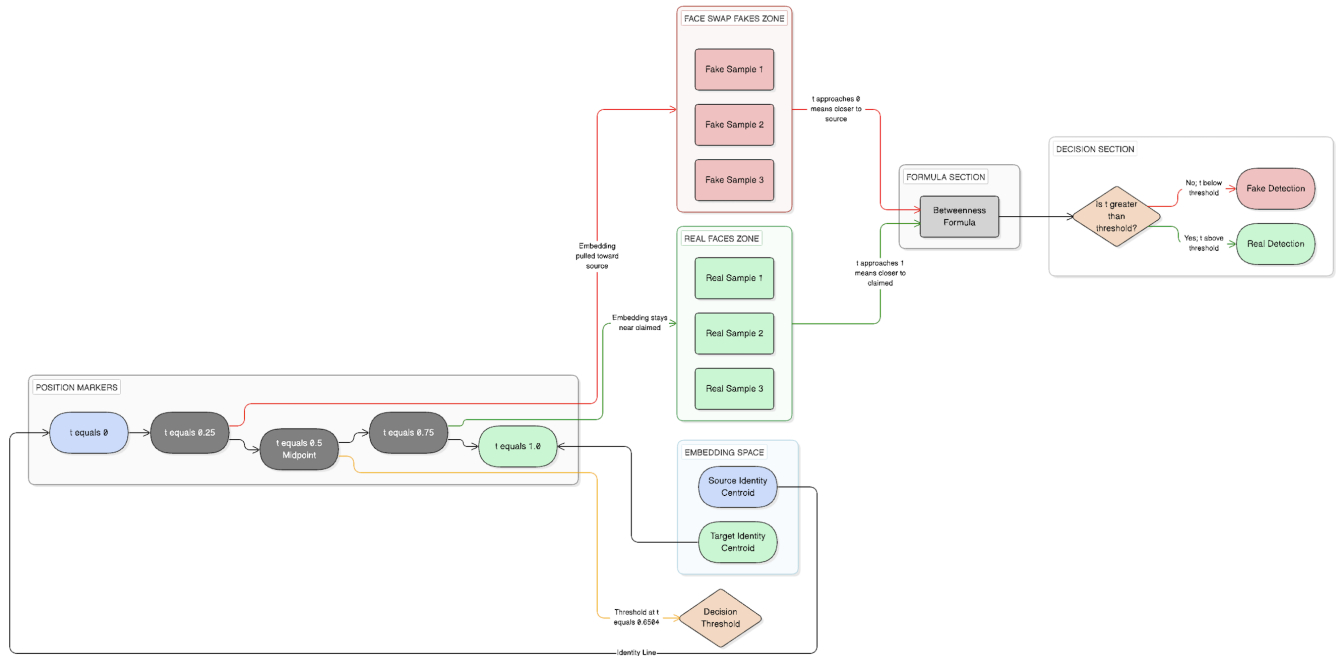


FIGURE 3. Inter-identity betweenness geometry. The identity line connects the source identity centroid ($t = 0$) to the target identity centroid ($t = 1.0$), with position markers at $t = 0.25, 0.5$ (midpoint), and 0.75 . Face-swap fakes fall in the red zone ($t \approx 0.38\text{--}0.48$) because their embeddings get pulled toward the source identity during the swapping process. Real faces stay in the green zone ($t \approx 0.84\text{--}0.92$) near the claimed identity. The betweenness formula computes t for each test embedding and the decision section applies the threshold ($\tau = 0.6504$ on Celeb-DF v2) to classify the face as real or fake.

just validating our model's theory this also opens up a room for practical use cases. When we look at the situations where one actually needs to trace back whose face was used as the donor, our model's geometric framework gives you a reasonable way to do that.

H. RELATIONSHIP BETWEEN METHODS

Scoring methods, majorly four FACTOR, betweenness, margin and statistical are designed to capture different scenarios of how face swap deepfakes exists in embedding space. FACTOR goes for measuring proximity to the claimed identity. Betweenness comes into play by measuring position relative to two identities. Margin measures that gap between the best-matching and next best matching identities. Statistical modeling keeps tab regarding the consistency of similarity across the reference set. A Spearman rank correlation analysis (presented in detail in Section VI) confirms that these methods capture substantially non-redundant information. The lowest correlation is between betweenness and FACTOR ($\rho = 0.729$), indicating that betweenness captures a meaningful proportion of unique variance not present in the baseline. The highest is between margin and betweenness ($\rho = 0.930$), which is expected because both methods inherently involve comparison across identities, though the specific computation differs. We also evaluate a simple ensemble combination (M+S+B: margin + statistical + betweenness, averaged) that achieves 99.99% on Celeb-DF and 99.94% on FaceForensics++ Deepfakes. However, since the standalone betweenness method already achieves

99.99% on Celeb-DF and 99.97% on FaceForensics++, the ensemble does not improve upon the best individual method and in fact marginally reduces FF++ performance due to dilution from margin's lower FaceForensics++ scores. We report the ensemble for completeness but recommend standalone betweenness as the primary method for deployment.

I. VIDEO-LEVEL AGGREGATION AND DECISION

Regardless of which scoring method is used, the per-frame scores are aggregated to a single video-level score via arithmetic mean:

$$S_{\text{video}} = \frac{1}{N} \sum_{i=1}^N S(e_i)$$

where N is the number of successfully processed frames (up to 32; formalized as Equation 6 in Section IV). The video is classified as real if $S_{\text{video}} \geq \tau$ and as fake otherwise, where the threshold τ is determined by Youden's J statistic [62] on a held out development set. Youden's J maximizes the given sum of sensitivity and does the minus one which provides an optimal operating point. This is for balancing the cost of false positives against false negatives. The threshold depends on the dataset i.e for betweenness scoring, $\tau = 0.6504$ on Celeb-DF v2 and different thresholds apply to FaceForensics++ conditions. This calibration requirement is a practical constraint we discuss honestly in Section VII. Simple arithmetic averaging was chosen after testing more complex aggregation schemes, including bounding-box-weighted averaging (weighting frames by detection confidence), me-

dian aggregation, and trimmed means. None provided any improvement over simple averaging, a finding consistent with the observation that our per-frame signals are already strong and consistent. The formal decision rule and complete mathematical formulation are presented in Section IV.

IV. THEORETICAL BACKGROUND AND MATHEMATICAL FORMULATION

This section formalizes the scoring functions and decision procedures introduced conceptually in Section III. We present the complete mathematical framework, beginning with notation and the fundamental similarity measure, proceeding through each of the four scoring functions, and concluding with the video-level aggregation and decision rule. Three algorithm pseudocodes are provided to give a self-contained, reproducible specification of the entire detection pipeline.

A. NOTATION AND PRELIMINARIES

Let $\mathbf{e} \in \mathbb{R}^{512}$ denote the L2-normalized embedding vector produced by the frozen ArcFace ResNet-50 model for a given face crop. For the claimed identity c , we denote the reference embedding set as $\mathcal{R}_c = \{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_M\}$, where M is the number of reference embeddings (approximately 160 for Celeb-DF v2, 16 for FaceForensics++). The centroid of the claimed identity is computed as:

$$\mathbf{c}_{\text{claimed}} = \frac{1}{M} \sum_{i=1}^M \mathbf{r}_i \quad (1)$$

Similarly, for every other identity $j \neq c$ in the reference database, we precompute centroids $\mathbf{c}_j$. Given a test video V , we extract N face embeddings $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_N\}$ from the N successfully detected frames (with $N \leq 32$, as described in Section III-B).

B. COSINE SIMILARITY

All scoring functions in our framework are built upon cosine similarity, which measures the angular proximity between two embedding vectors. Since ArcFace produces L2-normalized embeddings that lie on a unit hypersphere, cosine similarity reduces to the dot product:

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \mathbf{a} \cdot \mathbf{b} \quad (2)$$

The second equality holds because $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$ for L2-normalized vectors. The resulting similarity value lies in $[-1, 1]$, where values close to 1 indicate high identity similarity and values near 0 or below indicate different identities. In practice, cosine similarities between embeddings of the same person typically range from 0.5 to 0.9, while similarities between different people are usually below 0.3.

C. SCORING FUNCTIONS

We now formalize the four scoring functions, each of which assigns a per-frame score to a test embedding $\mathbf{e}$.

1) FACTOR Score (Baseline)

The FACTOR baseline computes the maximum cosine similarity between the test embedding and all reference embeddings for the claimed identity:

$$S_{\text{FACTOR}}(\mathbf{e}) = \max_{\mathbf{r} \in \mathcal{R}_c} \text{sim}(\mathbf{e}, \mathbf{r}) \quad (3)$$

This score answers a single question: how closely does the test face match the best-matching reference of the claimed identity? Real faces produce high scores because at least one reference embedding will be closely aligned. Face-swap fakes produce lower scores because their embeddings have been displaced from the claimed identity's cluster. The FACTOR score captures intra-identity similarity but ignores all inter-identity relationships.

2) Negative Margin Score

The negative margin score extends the FACTOR principle by introducing a comparative element. It measures the gap between the maximum similarity to the claimed identity and the maximum similarity to the closest competing identities:

$$S_{\text{margin}}(\mathbf{e}) = \max_{\mathbf{r} \in \mathcal{R}_c} \text{sim}(\mathbf{e}, \mathbf{r}) - \max_{j \in \text{top-}k, j \neq c} \max_{\mathbf{r} \in \mathcal{R}_j} \text{sim}(\mathbf{e}, \mathbf{r}) \quad (4)$$

where top- k denotes the k identities (excluding c) whose centroids are nearest to $\mathbf{e}$. We use $k = 5$ in all experiments. In the case of real face, the first term is large while the second term is small finally producing a greater positive margin. Face swap fakes residual source identity features tend to increase this second term parallelly the imperfect identity transfer leans towards reducing the first, thus shrinking the margin. The sign and magnitude of this margin serve as the detection signal.

3) Betweenness Parameter

The betweenness parameter captures where a test embedding sits on the geometric line connecting two identity centroids. We first identify the nearest other identity as the non-claimed identity with the highest centroid similarity to $\mathbf{e}$:

$$\mathbf{c}_{\text{other}} = \mathbf{c}_{j^*}, \quad \text{where } j^* = \underset{j \neq c}{\text{argmax}} \text{sim}(\mathbf{e}, \mathbf{c}_j) \quad (5)$$

The betweenness parameter is then computed as:

$$t(\mathbf{e}) = \frac{\text{sim}(\mathbf{e}, \mathbf{c}_{\text{claimed}})}{\text{sim}(\mathbf{e}, \mathbf{c}_{\text{claimed}}) + \text{sim}(\mathbf{e}, \mathbf{c}_{\text{other}})} \quad (6)$$

This parameter ranges from 0 to 1. When $t = 1$, the embedding is fully aligned with the claimed identity (zero similarity to the other). When $t = 0.5$, the embedding sits at the exact midpoint between two identities, the signature of a perfect face-swap blend. In our experiments, real faces yield $t \approx 0.84$ – 0.92 and face-swap fakes yield $t \approx 0.38$ – 0.48 . The parameter t itself is used directly as the detection score: higher values indicate a real face.

4) Statistical Score

The statistical score replaces the maximum operation in FACTOR with a mean over all reference embeddings, capturing overall consistency rather than best-case alignment:

$$S_{\text{stat}}(\mathbf{e}) = \frac{1}{|\mathcal{R}_c|} \sum_{\mathbf{r}_i \in \mathcal{R}_c} \text{sim}(\mathbf{e}, \mathbf{r}_i) \quad (7)$$

A genuine face of the claimed identity will be consistently similar to all reference embeddings, because natural within-person variation in ArcFace space is small relative to between-person distances. A face-swap fake may achieve high similarity to a few references by chance, but its mean similarity across the full reference set will be depressed by the systematic misalignment that imperfect identity transfer creates.

D. VIDEO-LEVEL AGGREGATION

Each scoring function produces a per-frame score. To arrive at a single video-level decision, we aggregate across all N successfully processed frames using the arithmetic mean:

$$S_{\text{video}} = \frac{1}{N} \sum_{i=1}^N S(\mathbf{e}_i) \quad (8)$$

where $S(\cdot)$ is any of the four scoring functions defined above. We chose the arithmetic mean after evaluating several alternatives, including median aggregation, trimmed means (discarding the top and bottom 10% of frame scores), and bounding-box-weighted averaging (weighting each frame by its face detection confidence). None of these alternatives improved upon simple averaging, which we attribute to the fact that our per-frame scores are already strong and exhibit low variance within a given video. There is an alluring feature of mean too that is, it is sensitive to outliers frames that tend to signal partial or intermittent face swapping.

E. DECISION RULE

By comparing the video level score relative to the threshold the final classification is made. τ :

$$\hat{y} = \begin{cases} \text{Real,} & \text{if } S_{\text{video}} \geq \tau \\ \text{Fake,} & \text{if } S_{\text{video}} < \tau \end{cases} \quad (9)$$

Youden's J statistic [62] is used for finding out the threshold τ which is then used to select the operating point that tends to maximize the sum of sensitivity i.e true positive rate and specificity the true negative rate minus one:

$$\tau^* = \underset{\tau}{\operatorname{argmax}} [\text{TPR}(\tau) + \text{TNR}(\tau) - 1] \quad (10)$$

A principled, data driven threshold is obtained through this criterion that balances between the cost of false positives that are real videos flagged as fake and against false negatives that is fake videos passing as real. The threshold is computed on a development set and is inherently dataset-dependent. For example, the optimal betweenness threshold on Celeb-DF v2 is $\tau = 0.6504$, while on FaceForensics++

Algorithm 1 Complete Detection Pipeline

Require: Test video V , reference database $\mathcal{D} = \{(\mathcal{R}_j, \mathbf{c}_j)\}$ for all identities j , claimed identity c , threshold τ , scoring function $S(\cdot)$

Ensure: Label $\hat{y} \in \{\text{Real}, \text{Fake}\}$, confidence score

```

1: Sample 32 frames uniformly from  $V$ 
2: Initialize score list  $\mathcal{S} \leftarrow []$ 
3: for each sampled frame  $I_i$  do
4:   faces  $\leftarrow$  RetinaFace( $I_i$ ) {detect faces}
5:   if no face detected then
6:     continue {skip frame}
7:   end if
8:   crop  $\leftarrow$  Align(faces[0],  $112 \times 112$ ) {5-point alignment}
9:    $\mathbf{e}_i \leftarrow$  ArcFace(crop) {512-dim, L2-normalized}
10:   $s_i \leftarrow S(\mathbf{e}_i, \mathcal{D}, c)$  {any of Eq. 2-5}
11:  Append  $s_i$  to  $\mathcal{S}$ 
12: end for
13:  $S_{\text{video}} \leftarrow \text{mean}(\mathcal{S})$  {Eq. 6}
14: conf  $\leftarrow |S_{\text{video}} - \tau|$ 
15: if  $S_{\text{video}} \geq \tau$  then
16:   return Real, conf
17: else
18:   return Fake, conf
19: end if
```

Deepfakes it differs due to the different score distributions that arise from having 1,000 identities with 16 reference frames each versus 59 identities with approximately 160 references each. This calibration requirement is an honest practical constraint of our approach, discussed further in Section VII.A confidence score is also provided alongside the binary decision, defined as the absolute distance from the threshold: $\text{conf} = |S_{\text{video}} - \tau|$. Larger values indicate higher confidence in the classification, whether real or fake.

F. ALGORITHM PSEUDOCODES

We present three algorithms that together provide a complete, self-contained specification of the detection pipeline. Algorithm 1 describes the full end-to-end process from raw video input to a real/fake decision. Algorithm 2 details the per-frame betweenness geometry computation, which is the most involved scoring function. Algorithm 3 specifies the video-level aggregation and threshold-based decision. A few implementation notes are worth mentioning. In Algorithm 1, we process only the first detected face per frame; in both Celeb-DF and FaceForensics++, videos contain a single subject, so this is sufficient. In datasets with multiple faces per frame, the face corresponding to the claimed identity would need to be selected, for instance by choosing the detection with the highest similarity to the claimed centroid. In Algorithm 2, the loop over all other identities can be accelerated using approximate nearest-neighbor structures if the reference database is very large, though for the database sizes in our experiments (59 and 1,000 identities),

Algorithm 2 Betweenness Geometry Scoring (Per-Frame)**Require:** Test embedding $\mathbf{e}$, claimed centroid $\mathbf{c}_{\text{claimed}}$, all other centroids $\{\mathbf{c}_j\}_{j \neq c}$ **Ensure:** Betweenness parameter $t \in [0, 1]$

```

1:  $s_{\text{claimed}} \leftarrow \text{sim}(\mathbf{e}, \mathbf{c}_{\text{claimed}})$ 
2: {Find nearest non-claimed identity}
3:  $s_{\text{best}} \leftarrow -\infty$ 
4: for each identity  $j \neq c$  do
5:    $s_j \leftarrow \text{sim}(\mathbf{e}, \mathbf{c}_j)$ 
6:   if  $s_j > s_{\text{best}}$  then
7:      $s_{\text{best}} \leftarrow s_j$ 
8:   end if
9: end for
10:  $s_{\text{other}} \leftarrow s_{\text{best}}$ 
11: {Compute betweenness parameter}
12:  $t \leftarrow s_{\text{claimed}} / (s_{\text{claimed}} + s_{\text{other}})$  {Eq. 4}
13: return  $t$  {high  $t$  = likely real, low  $t$  = likely fake}

```

Algorithm 3 Video-Level Aggregation and Decision**Require:** Frame scores $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$, threshold τ **Ensure:** Label $\hat{y}$, confidence score

```

1:  $S_{\text{video}} \leftarrow \frac{1}{N} \sum_{i=1}^N s_i$  {arithmetic mean, Eq. 6}
2:  $\text{conf} \leftarrow |S_{\text{video}} - \tau|$ 
3: if  $S_{\text{video}} \geq \tau$  then
4:    $\hat{y} \leftarrow \text{Real}$ 
5: else
6:    $\hat{y} \leftarrow \text{Fake}$ 
7: end if
8: return  $\hat{y}$ ,  $\text{conf}$ 

```

brute-force search completes in under 0.1 milliseconds per frame. The computational overhead of all scoring functions combined is less than 0.03% of the total pipeline time, as the dominant cost is face detection and embedding extraction (detailed in Section VII).

V. EXPERIMENTAL SETUP

This section documents the datasets, split protocols, hardware and software environment, and evaluation metrics used throughout our experiments. Each step is presented in detail for full reproducibility.

A. DATASETS AND SPLIT PROTOCOLS

We evaluate on two standard benchmarks spanning six distinct test conditions, as summarized in Table 2 (Section II). The split protocols differ between datasets to reflect their respective structures, and we describe each in detail below.

Celeb-DF v2. This dataset contains 590 real videos and 5,639 deepfake videos spanning 59 celebrity identities. We follow a subject-disjoint 50/50 split: half of the identities are assigned to a development set (for threshold calibration) and the other half to the test set. No identity appears in both splits, preventing any identity-level information leakage. For each identity, reference embeddings are extracted from

the real videos of that person, yielding approximately 160 reference embeddings per identity after processing at 32 frames per video. All deepfake videos targeting that identity serve as the fake test set, while the real videos not used for reference serve as the real test set. We verified that no reference frame overlaps with any test frame by ensuring that reference and test frames are drawn from separate video files.

FaceForensics++ (all conditions). This dataset provides 1,000 original YouTube videos, each manipulated by four methods: Deepfakes, FaceSwap, Face2Face, and NeuralTextures. Each original video features a single identity, giving 1,000 identities total. We follow a temporal split within each video: the first 16 frames serve as reference embeddings, and the last 16 frames serve as test embeddings. This protocol ensures zero temporal overlap between reference and test data. The same reference set (from the original, unmanipulated videos) is used across all four manipulation conditions, which mirrors the realistic scenario where reference material comes from authentic footage. For the compression robustness evaluation (FF++ Deepfakes c40), reference embeddings are extracted from the c23 (light compression) versions of the original videos, while test embeddings come from the c40 (heavy compression) versions of both original and manipulated videos. This deliberately mismatched setup simulates the practical scenario where reference material is higher quality than the test video, as would be typical when comparing a social media upload against enrollment footage.

Data leakage verification. We explicitly verified that no data leakage exists in any experimental condition. On Celeb-DF v2, the subject-disjoint split guarantees identity-level isolation. On FaceForensics++, the temporal split within each video guarantees frame-level isolation. Furthermore, our scoring functions use only test embeddings and pre-computed reference data (centroids and stored embeddings); no information from the test set influences the reference database or threshold calibration. Table 3 summarizes the split protocols and key parameters for all six evaluation conditions.

B. IMPLEMENTATION DETAILS

Hardware. All experiments were conducted on a single workstation running Windows with an NVIDIA RTX 5070Ti GPU. No distributed computing or cloud resources were used, underscoring the computational accessibility of our approach.

Software. The implementation is in Python 3.11, using ONNX Runtime for ArcFace inference, InsightFace for face detection and alignment, NumPy and scikit-learn for similarity computation and metric calculation, and Matplotlib for figure generation. The entire pipeline runs in a Jupyter Notebook environment. No deep learning training framework (PyTorch, TensorFlow) is needed at inference time, since the ArcFace model is loaded as a frozen ONNX graph.

TABLE 3. Dataset split protocols and experimental parameters for all six evaluation conditions. Leakage prevention mechanisms are documented for each condition.

Condition	Real	Fake	IDs	Ref/ID	Test Fr.	Split Method & Leakage Prevention
Celeb-DF v2	590	5,639	59	~160	32	Subject-disjoint 50/50; separate video files
FF++ Deepfakes c23	1,000	1,000	1,000	16	16	Temporal split; first 16 ref, last 16 test
FF++ FaceSwap c23	1,000	1,000	1,000	16	16	Same refs as Deepfakes; different fakes
FF++ Face2Face c23	1,000	1,000	1,000	16	16	Same refs; reenactment (negative control)
FF++ NeuralTex c23	1,000	1,000	1,000	16	16	Same refs; reenactment (negative control)
FF++ Deepfakes c40	1,000	1,000	1,000	16	16	Refs from c23, test from c40 (quality mismatch)

Face detection and embedding. InsightFace RetinaFace is used for face detection with default confidence thresholds. Detected faces are aligned to 112×112 pixels using the standard InsightFace five-point alignment. The ArcFace ResNet-50 model (pretrained on WebFace600K, 512-dimensional output, L2-normalized) is used without any modification.

Frame sampling. We uniformly sample 32 frames per video. For Celeb-DF v2, frames are sampled across the full video duration. For FaceForensics++, the last 16 frames of each video are used as test frames (with the first 16 reserved as reference). Frames where face detection fails are skipped; this occurs in fewer than 1% of frames across both benchmarks.

Scoring parameters. The negative margin method uses $k = 5$ nearest other identities. The betweenness method selects the single nearest other identity per frame. All scoring computations use cosine similarity on L2-normalized embeddings, which reduces to a dot product.

C. EVALUATION METRICS

Primary metric: Video-level AUC. Our primary evaluation metric is the Area Under the Receiver Operating Characteristic Curve (AUC), computed at the video level. Each video receives a single score (the arithmetic mean of per-frame scores), and the AUC is computed over all videos in the test set. Threshold independent metric called AUC is used and provides a relative comprehensive measure of discriminative ability across all possible available operating points.

Bootstrap confidence intervals: To quantify the statistical reliability of our AUC estimates, we compute 95% confidence intervals using bootstrap resampling with 1,000 iterations. In each iteration, we resample the test set with replacement and compute the AUC, yielding a distribution from which the 2.5th and 97.5th percentiles define the confidence interval.

Threshold selection. For computing binary classification metrics (accuracy, precision, recall, F1, specificity), we determine the threshold using Youden's J statistic on the development set, as described in Section IV. This threshold is then applied to the test set without further adjustment.

Frame-level AUC. As a secondary metric, we also report frame-level AUC on FaceForensics++ (all manipulations pooled), treating each frame as an independent sample. This confirms that the detection signal exists at the individual

frame level and is not solely an artifact of video-level aggregation.

Additional metrics. For the comparison with published methods (Section VI-B), we report accuracy, precision, recall, F1, and specificity at the Youden-optimal threshold. These are provided to enable direct comparison with methods that report classification metrics rather than AUC.

VI. RESULTS AND ANALYSIS

This section presents our experimental findings across six evaluation conditions. We begin with the main cross-dataset results and statistical significance analysis, then compare against published methods. We follow with manipulation breakdowns, compression robustness, ablation studies (reference set size, method independence, and failed methods), score distribution visualizations, embedding space analysis, source identity detection validation, failure case examination, and frame-level versus video-level comparisons.

A. MAIN RESULTS

Table 4 presents video-level AUC (%) for all five scoring methods across all six evaluation conditions. Several patterns are immediately apparent. On face swap datasets, all identity-based methods perform exceptionally well. Betweenness achieves the highest AUC on every face-swap condition: 99.99% on Celeb-DF v2, 99.97% on FF++ Deepfakes, 99.92% on FF++ FaceSwap, and 99.94% on FF++ Deepfakes c40. The improvement over the FACTOR baseline is most pronounced on Celeb-DF v2 (+0.59 percentage points), where the larger reference sets (approximately 160 embeddings per identity) allow the geometric methods to exploit inter-identity structure more effectively. On reenactment datasets (Face2Face and NeuralTextures), all methods perform at or near chance level. FACTOR achieves the highest AUC on Face2Face at 88.78%, which may seem surprisingly high for an identity-based method on identity-preserving manipulations. We attribute this to Face2Face's expression transfer occasionally introducing subtle identity drift that FACTOR's max-similarity scoring can detect. Our geometric methods, which are specifically tuned to detect inter-identity interpolation, show lower performance on Face2Face (71.50% to 82.69%) because the manipulation does not create the between-identity signature they are designed to find. Methods tend to stay around 50% making

TABLE 4. Video level AUC (%) presented across all methods and evaluation conditions that are present. Best result per column is given in **bold**. Face swap conditions for Celeb-DF, FF++ DF, FF++ FS, FF++ c40 show about near perfect performance score. Reenactment conditions on FF++ F2F, FF++ NT confirms the boundary expected scope of identity based methods.

Method	Celeb-DF	FF++ DF	FF++ FS	FF++ F2F	FF++ NT	FF++ c40
FACTOR	99.40	99.88	99.87	88.78	46.18	99.83
Statistical	99.80	99.85	99.83	81.94	50.37	99.81
Margin	99.90	99.82	99.06	71.50	48.05	99.36
Betweenness	99.99	99.97	99.92	82.69	48.18	99.94
M+S+B (ensemble)	99.99	99.94	99.90	75.78	49.17	99.91

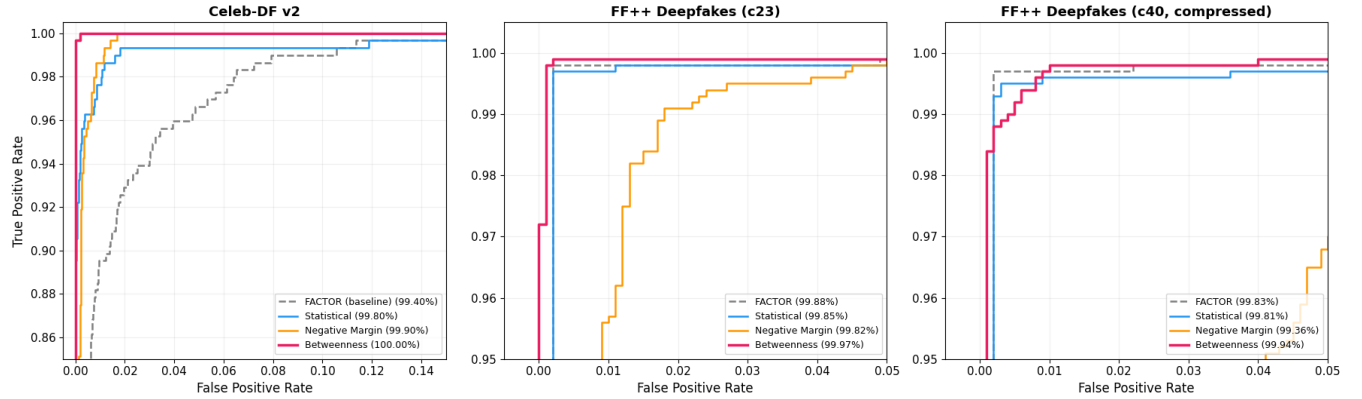


FIGURE 4. ROC curves given according to all four scoring methods on three evaluation conditions: First Celeb-DF v2 (left), FF++ Deepfakes c23 (center), and FF++ Deepfakes c40 (right) dataset are used. Betweenness (pink, thickest) dominates all panels throughout. On Celeb-DF v2 the gap present between betweenness and FACTOR (gray, dashed line) is most visible at low false positive rates. On FF++ dataset conditions all methods generally perform well and with betweenness maintaining the tightest curve.

it a random chance on NeuralTexture. This finding confirms that this texture based reenactment often leaves the identity signal itself completely intact. These texture based reenactment results are not termed as failures of our method. one can say they are more like confirmations of scope. Identity based detection is fundamentally appropriate to be used for identity replacing manipulations and we report these reenactment results in order to establish where the boundary lies. Table 5 gives bootstrap 95% confidence intervals for the primary method on the key benchmarks dataset. The betweenness technique achieves to a confidence interval of [100.00%, 100.00%] on Celeb-DF version 2 implying that across 1,000 bootstrap resamples there is not even a single resample that produced an AUC below 100.00% after rounded to two decimal places. On FF++ Deepfakes and FF++ c40 these confidence intervals are similarly tight assuring that any high AUC values are statistically robust and not just any artifacts of particular test set compositions. Fig. 4 shows the ROC curves for all four present methods and that to on three key conditions that are: Celeb-DF v2, FF++ Deepfakes c23, and FF++ Deepfakes c40 dataset. The betweenness curve with colour code pink and thickest line dominates all panels and aligns for reaching a perfect or near perfect true positive rates along with extremely low false positive rates.

B. COMPARISON WITH PUBLISHED METHODS

Table 6 puts out our results in context against the published detection methods that are evaluated on the same benchmarks. The comparison goes on by using Celeb-DF v2 AUC as the primary metric. This is preferred because this is where cross dataset generalization is most transparently presented. Many worthy highlights are observed during this. First is the gap between our betweenness method (99.99%) and the best available supervised cross dataset result of Face X-Ray at 93.47% and this is over 6 percentage points. At some point this is not an incremental improvement, it gives off a reflection of fundamentally different relationship to the generalization problem. Second point is that our method achieves this accuracy without any training on deepfake data itself. On the contrary Face X-Ray and SBI both require training pipelines along with self generated pseudo forgeries. Third and last is that majority of supervised methods tend to cluster between 55% and 85% on Celeb-DF which can be called range where generalization has historically been the hardest challenge to world. There are two caveats which we counter regarding this comparison. First one comes up as that different methods use different evaluation protocols e.g. train/test splits, pre processing thus the numbers are not perfectly aligned. Our report used only the values presented in the original publications and the standardized numbers from the DeepfakeBench framework

TABLE 5. Bootstrap 95% confidence intervals containing 1,000 resamples used for video level AUC on key benchmark dataset.

Dataset	Method	AUC (%)	95% CI
Celeb-DF	Betweenness	99.99	[100.00, 100.00]
Celeb-DF	FACTOR	99.40	[99.16, 99.61]
FF++ DF c23	Betweenness	99.97	[99.91, 100.00]
FF++ DF c40	Betweenness	99.94	[99.83, 99.99]

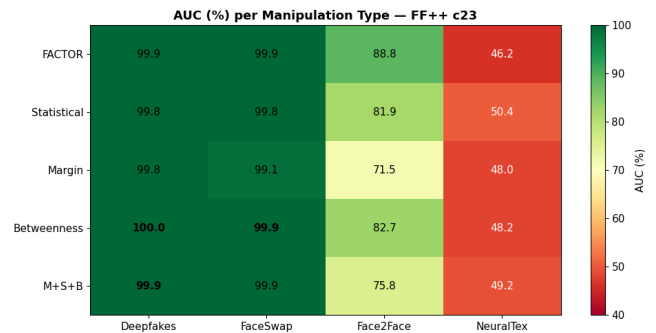
TABLE 6. Comparison relative to the published methods on Celeb-DF v2 (AUC %). Methods are grouped by genre for better understanding. Our betweenness method achieves the highest reported AUC out of all with zero deepfake training.

Method	Celeb-DF (%)	Training
MesoNet [11]	54.8	Supervised
Capsule-Forensics [43]	57.5	Supervised
XceptionNet [12]	65.3	Supervised
FakeSpotter [50]	66.8	Supervised
Multi-Attentional [42]	67.02	Supervised
RECCE [46]	68.71	Supervised
SPSL [41]	76.9	Supervised
EfficientNet-B3 [55]	85.0	Supervised
TALL-Swin [45]	90.79	Supervised
Thinking in Freq. [14]	91.0	Supervised
SBI [19]	93.18	Self-gen.
Face X-Ray [18]	93.47	Self-gen.
FACTOR [21]	99.40	None
Ours (Betweenness)	99.99	None

[49]. Second point is that our method requires a authentic reference database for the claimed identity verification which supervised methods do not use. This is a notable constraint that limits our model applicability to scenarios where reference material should exist, as discussed in Section VII.

C. PER-MANIPULATION ANALYSIS

Fig. 5 shows a heatmap of the AUC values for all the five methods and throughout four FaceForensics++ manipulation types. The visual pattern that comes out is striking: the face swap columns of Deepfakes and FaceSwap are uniformly green (99%+), while on the reenactment columns of Face2Face, NeuralTextures range goes from yellow to red. Within these face swap columns our betweenness technique consistently came out as the top performer, though the differences are considerably small at this performance level. The more interesting variation for us appeared in the reenactment columns. FACTOR presents the highest Face2Face performance (88.78%) and this is likely because its max similarity scoring goes on by capturing the subtle identity drift that happens when expression transfer is introduced. Our margin method shows the least Face2Face performance (71.50%) but it made sense as margin scoring

**FIGURE 5.** AUC (%) depicted per manipulation type used on FaceForensics++ c23. Face swap methods like Deepfakes, FaceSwap show near perfect detection score across all scoring methods tested. Reenactment methods such as Face2Face, NeuralTextures also confirm the scope boundary of identity-based detection.

explicitly looks for competing identity signals in dataset and reenactment does not introduce a competing identity itself. The NeuralTextures column is essentially introduced to be random for all the tested methods, with AUC values going between 46% to 50%. This is the another expected result as NeuralTextures modifies only the mouth region of individual texture rendering without any altering identity this there is simply no identity based signal to detect in media.

D. COMPRESSION ROBUSTNESS

Real world video typically undergoes great amount of lossy compression during distribution making compression robustness a critical parameter for practical development. Table 7 and Fig. 6 compares performance between light compression (c23) and heavy compression (c40) on FF++ Deepfakes dataset. Betweenness can be called out a compression robust method as it drops only 0.03 percentage points that is from 99.97% to 99.94% while testing on compressed media. In relative comparison FACTOR drops 0.05 points and statistical drops 0.04 points. The margin method that is used in our model shows the largest degradation of 0.46 points, from 99.82% to 99.36% which is still comparatively very high absolute performance but bring lights on concern that margin scoring is more sensitive to compression artifacts. This sensitivity is likely raised because accurate identification of the nearest competing identity requires a very finer geometric resolution than what the simple centroid based betweenness computation offers. Recall that as mentioned in our c40 evaluation the reference embeddings come from c23 which was a high quality set and test embeddings themselves come from c40 which was heavily compressed media set. This quality mismatch can be called for a realistic deployment scenario and the fact that betweenness degrades only 0.03% under this condition is a strong indicator for practical robustness of model.

E. ABLATION STUDIES

We performed ablation analysis for three constructed categories: reference set size sensitivity, method independence

TABLE 7. Compression robustness on FF++ Deepfakes presented as AUC (%) at c23 (light) vs. c40 (heavy) compression. The drop indicates the degradation and reference embeddings is used from c23 in both cases.

Method	c23 (%)	c40 (%)	Drop
FACTOR	99.88	99.83	0.05
Statistical	99.85	99.81	0.04
Margin	99.82	99.36	0.46
Betweenness	99.97	99.94	0.03
S+B	99.92	99.89	0.03
M+S+B	99.94	99.91	0.03

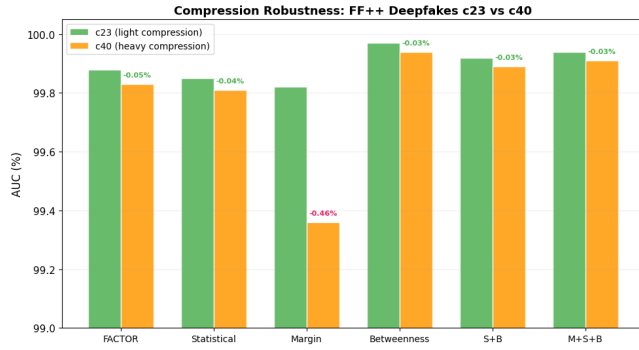


FIGURE 6. Compression robustness comparison. Green bars indicates c23 AUC, orange bars show c40 AUC. Percentage drop labeled above each of the c40 bar. Betweenness shows the smallest degradation out of all (0.03%), while margin shows the largest out there (0.46%).

verification and finally the documentation of failed methods and techniques.

1) Reference Set Size

A natural practical question is that comes up is how many of the reference frames per identity are needed for reliable detection score? Table 8 and Fig. 7 show AUC on Celeb-DF v2 with respect to the number of reference frames used varying from 4 to 128. One of the most striking analysis result is that betweenness achieves 99.98% AUC with just 4 reference frames per identity and even reaches 100.00% with 64 or more frames. This remarkable stability is proof that a single short video clip which yields roughly 4 usable face crops is sufficient for almost near perfect detection score. By contrast with the FACTOR model and statistical modeling show that they are more sensitivity to reference set size which goes for improving noticeably after 4 frames about 99.11% and 99.07% respectively to 128 frames giving 99.39% and 99.76%. The margin method can also be considered relatively stable across reference sizes difference, hovering between 99.89% and 99.94%. The practical implication of these finding is significant. For any deployment in video call authentication, any brief enrollment clip of a few seconds is enough to build a reference database that can supports up to 99.98% AUC detection. This low reference requirement largely broadens our range of scenarios where this method can be applied in real world.

TABLE 8. Reference set size ablation on Celeb-DF v2. AUC (%) as a function of reference frames per identity. Betweenness achieves 99.98% with just 4 frames.

Ref Frames	FACTOR	Margin	Statistical	Betweenness
4	99.11	99.91	99.07	99.98
8	99.22	99.91	99.19	99.98
16	99.14	99.90	99.17	99.97
32	99.00	99.89	99.10	99.97
64	99.41	99.94	99.73	100.00
128	99.39	99.89	99.76	100.00

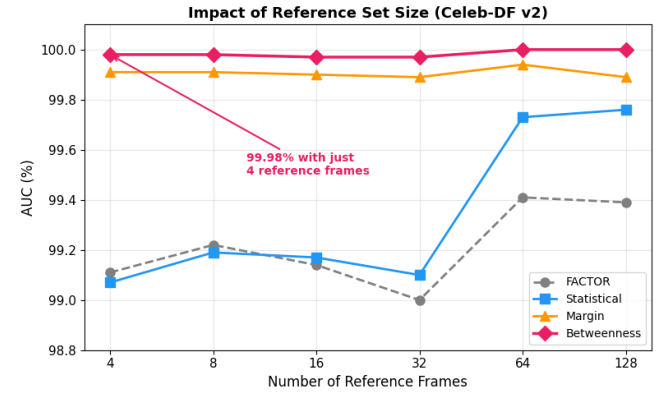


FIGURE 7. Impact of reference set size on dataset Celeb-DF v2. Betweenness represented by pink diamonds remains nearly flat at 99.98%+ regardless of what is the reference count, while FACTOR one with the gray circles and statistical with blue squares show relatively greater sensitivity to reference availability. The annotation highlights that 99.98% AUC is achievable with just 4 reference frames.

2) Method Independence

To verify that our three proposed methods capture genuinely different information from the FACTOR baseline (and from each other), we compute pairwise Spearman rank correlations between all four methods' video-level scores on Celeb-DF v2. Fig. 8 presents the correlation matrix. The key finding is that betweenness and FACTOR exhibit the lowest correlation ($\rho = 0.729$), confirming that betweenness captures a substantial proportion of unique variance not present in the baseline. If betweenness method was merely a rescaled version of FACTOR model, we would expect value of ρ near 1.0 but the observed value came out to be 0.729 demonstrating a genuine novelty in the signal being captured. Statistical modeling correlates this at $\rho = 0.853$ with FACTOR and $\rho = 0.785$ with betweenness model, indicating important but partial independence among them. Margin and betweenness on results show the highest correlation of $\rho = 0.930$, which was already anticipated because both methods inherently involve a comparison across the identities. However the correlation is still below 1.0, confirming that there is non redundancy. These correlation values are justifiable reporting as each of the involved method is tested individually rather than treating them as interchangeable and

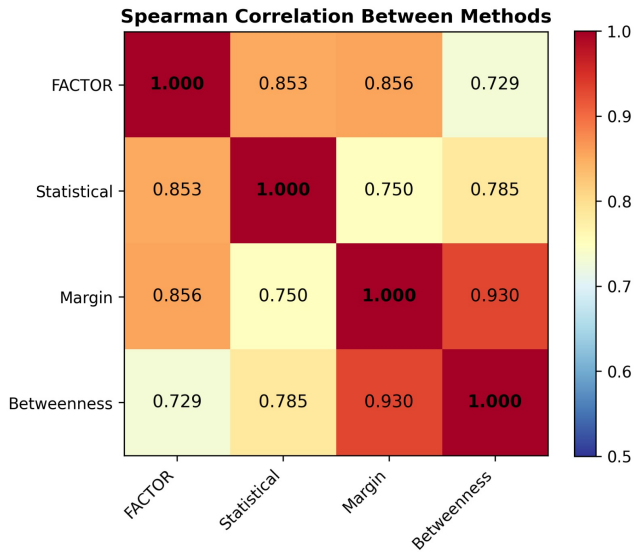


FIGURE 8. Spearman rank correlation matrix between all four scoring methods on Celeb-DF v2. Betweenness and FACTOR show the lowest correlation ($\rho = 0.729$), confirming that betweenness captures genuinely different information from the baseline. Darker colors indicate higher correlation.

they explain why the ensemble combination (M+S+B) does not substantially outperform standalone betweenness that is the margin signal is already largely captured by betweenness

3) Failed Methods

Scientific rigor demands that reporting not only what we worked on but also what did not follow through as intended. Table 9 documents all the four candidate methods that we implemented, evaluated but ultimately got rejected due to one or more reasons. We report them here both for transparency and because the failure modes are informative for future research needs. **Temporal Consistency Scoring (TCS)** aims to identify deepfakes by examining the variance of embeddings between frames. Although TCS achieved 64% AUC on the Celeb-DF v2 dataset, it dropped to 40% on the c40 subset of the dataset (indicating a worse result than random guesses). However, deepfakes produced today are temporally coherent almost alongside the real videos. This result correlates with existing literature on the topic. **Multi-layer feature fusion** aimed to fuse the embeddings of the early, middle and final layers of ArcFace. Intuitively one would expect that the early layers should be able to capture more fine grained texture information of the face that might not be visible to the final layer which is significantly focused on identifying faces itself. However, the best result of fusing the representations from the three different layers of ArcFace only achieved an AUC of 95.63% which is less than the performance of the final layer of ArcFace standalone (99.40%). The early layers of ArcFace encode generic information about the face that is mutually shared among all identities, which might later

TABLE 9. Failed methods: candidates that were implemented and evaluated but did not improve detection. Reporting these for scientific transparency.

Method	Best AUC	Why It Failed
Temporal Cons.	64% / 40%	Temporal coherence intact in modern fakes
Multi-Layer Fusion	95.63%	Early layers lack identity info; dilutes signal
Adaptive Thresh.	Hurts AUC	Adds noise to near-perfect separation
Frame Weighting	No gain	Uniform averaging already optimal

introduce noise when trying to recognize specific identities. **Adaptive thresholding** recalibrates the thresholds on a per identity basis, assuming that some of the identities have scores that are more clustered together than the others. Although this method is theoretically sound as thresholds is ranged from 0.035 to 0.732 but it hurts the AUC on both Celeb-DF and c40. Given that the baseline F_SCORE is already near perfect score, any introduced noise from recalibrating these thresholds on a per identity basis may lead to decrease in already excellent separating power of the model. **Frame weighting** assigns the weights to frames based on the confidence of the face detection algorithm in detecting a face in that particular frame itself. The hypothesis is that frames with higher confidence scores will have more reliable and accurate signals in determining the FACTOR score than ones with less confidence. As there is already sufficient signal between each of the present frame, the weighting algorithm does not provide any benefit to the model upon implementation. Fig. 9 depicts the FACTOR score (highest cosine similarity) distribution for real versus fake videos from each of the four datasets. For Celeb-DF v2, there is some overlap between the real and fake videos within the 0.55 to 0.65 similarity spectrum, which is the reason for the 99.40% AUC value of FACTOR for this dataset. For FF++ Deepfakes and FaceSwap datasets, the real and fake videos are more distinctly separated thus with the fake videos clustering at around 0.2 to 0.3 similarity scores and the real videos clustering at 0.85 to 0.95 similarity scores. Finally, for the c40 condition of the Deepfakes Million Video dataset, both the real and fake videos have broadened distributions due to the compression noise introduced during the video downloading process, but the gap between the two distributions remains preserved even after this. Fig. 10 shows the betweenness parameter t distributions for real versus fake videos from both the Celeb-DF v2 and FF++ Deepfakes datasets. The separation of the distributions is dramatic. For the Celeb-DF dataset, the real face examples cluster around a value of $t \approx 0.85$ and $t \approx 0.95$ (FF++), while fake faces cluster tends to be around $t \approx 0.45$ (Celeb-DF) and $t \approx 0.35$ (FF++). On both datasets, there can be observed a wide gap between the two distributions with no overlap virtually, explaining the almost perfect AUC scores. The dotted vertical lines at $t = 0.5$ for midpoint and $t = 1.0$

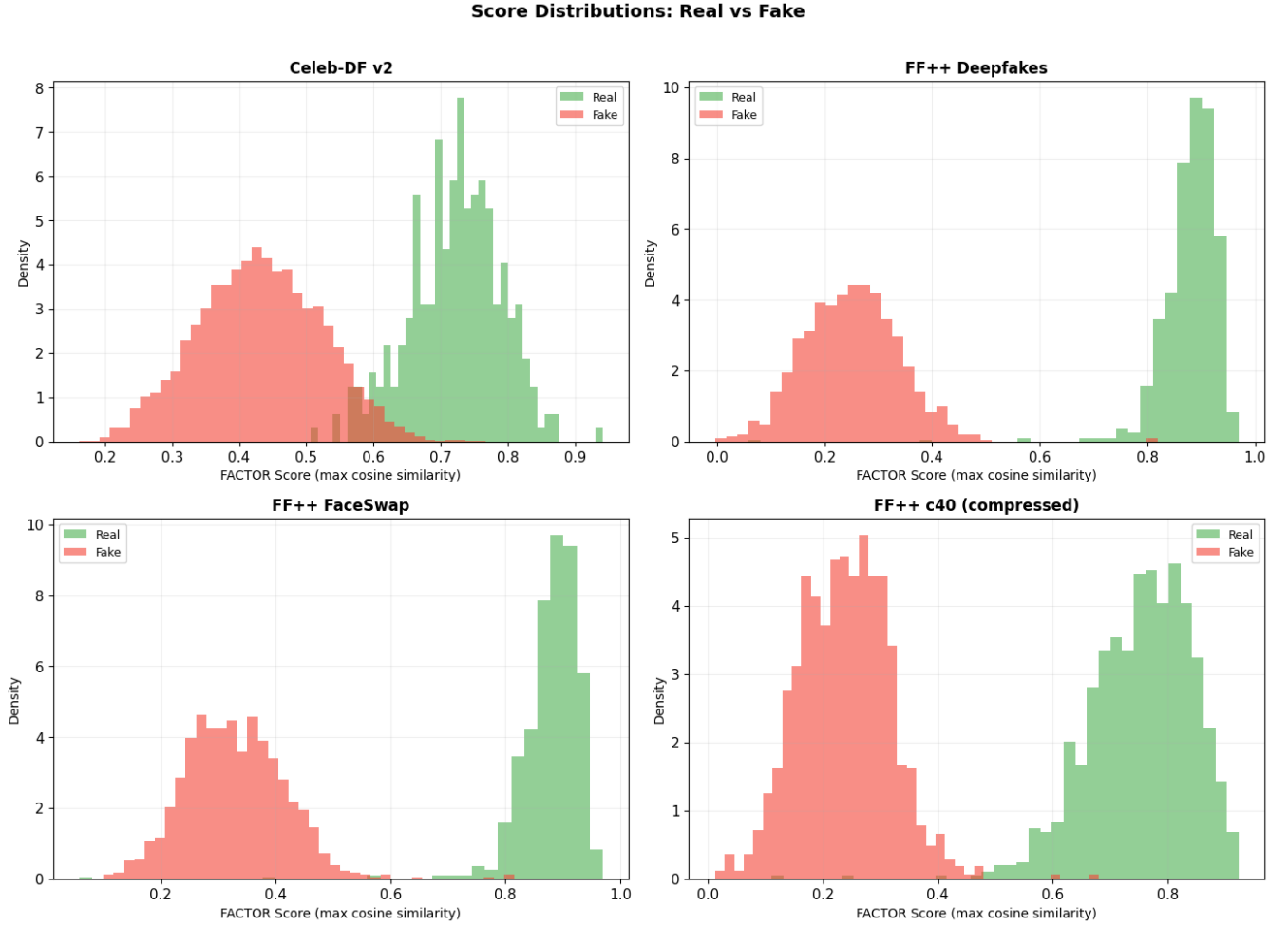


FIGURE 9. FACTOR score (max cosine similarity) distributions for real (green) and fake (red) videos across four datasets. Celeb-DF v2 (top left) shows the most overlap, consistent with it being the hardest benchmark. FF++ Deepfakes and FaceSwap (top right, bottom left) show clean separation. The c40 condition (bottom right) shows slight broadening from compression but maintains discriminative power.

for claimed identity provide visual anchors for the same.

F. EMBEDDING SPACE ANALYSIS

To further visualize the geometric structure our methods exploit, we present two complementary views of the embedding space. Fig. 11 shows a t-SNE projection of face embeddings from 10 randomly selected Celeb-DF identities. Real embeddings (circles) form tight, well-separated clusters for each identity, confirming that ArcFace produces highly discriminative representations. Fake embeddings (crosses) are scattered away from their target identity clusters and tend to appear in the spaces between clusters, visually confirming the inter-identity interpolation that our betweenness method is designed to detect. This t-SNE visualization provides an intuitive, low-dimensional confirmation of the geometric theory presented in Section III-D. Fig. 12 presents scatter plots relating betweenness t to FACTOR score on Celeb-DF v2 (left) and betweenness t to residual distance from the identity line on FF++ Deepfakes (right). On both panels, real (green) and fake (red) samples form clearly separable

clusters. On the Celeb-DF panel, the positive correlation between FACTOR score and betweenness t confirms that both signals are related, yet the spread within each cluster shows that betweenness captures additional variance not explained by FACTOR alone, which is consistent with their Spearman correlation of $\rho = 0.729$.

G. SOURCE IDENTITY DETECTION

As described in Section III-G, we validate our geometric interpolation theory by checking whether the nearest non-claimed identity centroid corresponds to the actual source (donor) identity used to create each deepfake. Table 10 and Fig. 13 present the results. The detection rates are consistently high: 96.2% on FF++ Deepfakes c23, 94.6% on Celeb-DF v2, 93.0% on FF++ FaceSwap c23, and 87.3% on FF++ Deepfakes c40. The FF++ Deepfakes rate of 96.2% is particularly significant: in nearly every fake video, the geometric structure of the embedding space correctly identifies which person's face was used as the donor. This is strong evidence that our betweenness method

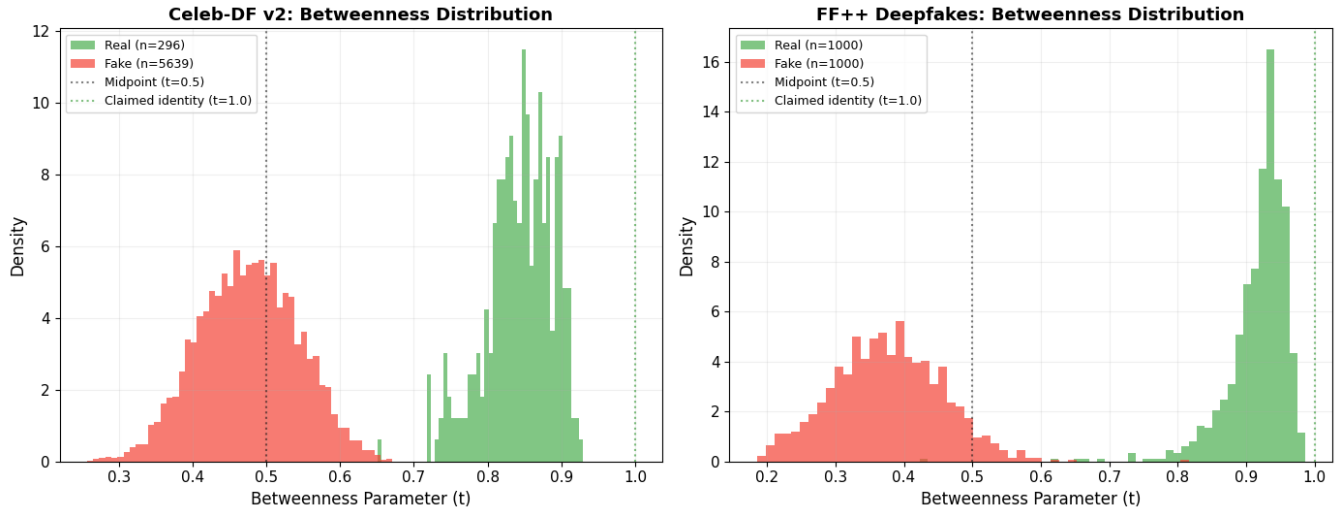


FIGURE 10. Betweenness parameter t distributions on Celeb-DF v2 (left) and FF++ Deepfakes (right). Real faces (green) cluster near the claimed identity ($t > 0.7$), while fake faces (red) cluster near the midpoint between identities ($t \approx 0.3$ – 0.5). The near-complete separation explains the 99.99% AUC. Dotted lines mark $t = 0.5$ (midpoint) and $t = 1.0$ (claimed identity).

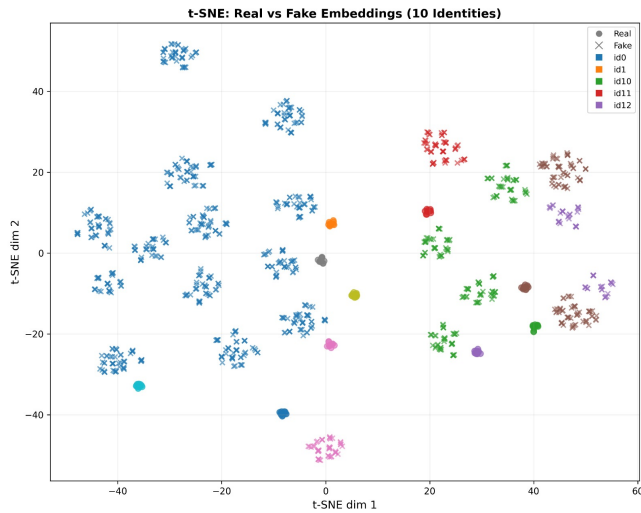


FIGURE 11. t-SNE visualization of ArcFace embeddings from 10 Celeb-DF identities. Circles represent real embeddings (tight clusters per identity), and crosses represent fake embeddings (displaced from target clusters, scattered between identities). This confirms the inter-identity interpolation that our betweenness method detects.

detects the actual face-swap mechanism, not coincidental statistical patterns. The variation across conditions is informative. The lower FaceSwap rate (93.0%) likely reflects the fact that FaceSwap’s 3D graphics-based approach creates a different interpolation geometry than autoencoder-based methods. The c40 rate (87.3%) shows that heavy compression degrades the fine geometric signal needed for source identification more than it degrades the binary real/fake decision (which only drops 0.03% in AUC). The Celeb-DF rate (94.6%) is slightly lower than FF++ Deepfakes despite the larger reference sets, possibly because Celeb-DF’s 59 identities are all celebrities whose faces may share

TABLE 10. Source identity detection rates across four evaluation conditions. High rates validate the geometric interpolation theory underlying our betweenness method.

Dataset	Detection Rate (%)
FF++ Deepfakes c23	96.2
Celeb-DF v2	94.6
FF++ FaceSwap c23	93.0
FF++ Deepfakes c40	87.3

more structural similarities than the more diverse FF++ population.

H. FAILURE CASE ANALYSIS

Understanding where a method fails is as informative as understanding where it succeeds. We examine the specific videos that our betweenness method misclassifies at the Youden-optimal threshold. **Celeb-DF v2.** At the optimal threshold of $\tau = 0.6504$, betweenness produces zero misclassified real videos (no authentic video is incorrectly flagged as fake) and only 11 missed fakes out of 5,639 deepfake videos (0.19% miss rate), where these 11 deepfakes pass the threshold and are incorrectly accepted as authentic. All 11 have betweenness scores in the narrow range of 0.65–0.67, just barely above the threshold. These represent exceptionally high-quality face swaps where the embedding sits very close to the claimed identity’s region in the space. The five hardest cases are:

- id32_id35_0006: $t = 0.6711$, FACTOR score 0.7349
- id34_id30_0001: $t = 0.6694$, FACTOR score 0.6556
- id24_id25_0007: $t = 0.6659$, FACTOR score 0.5545

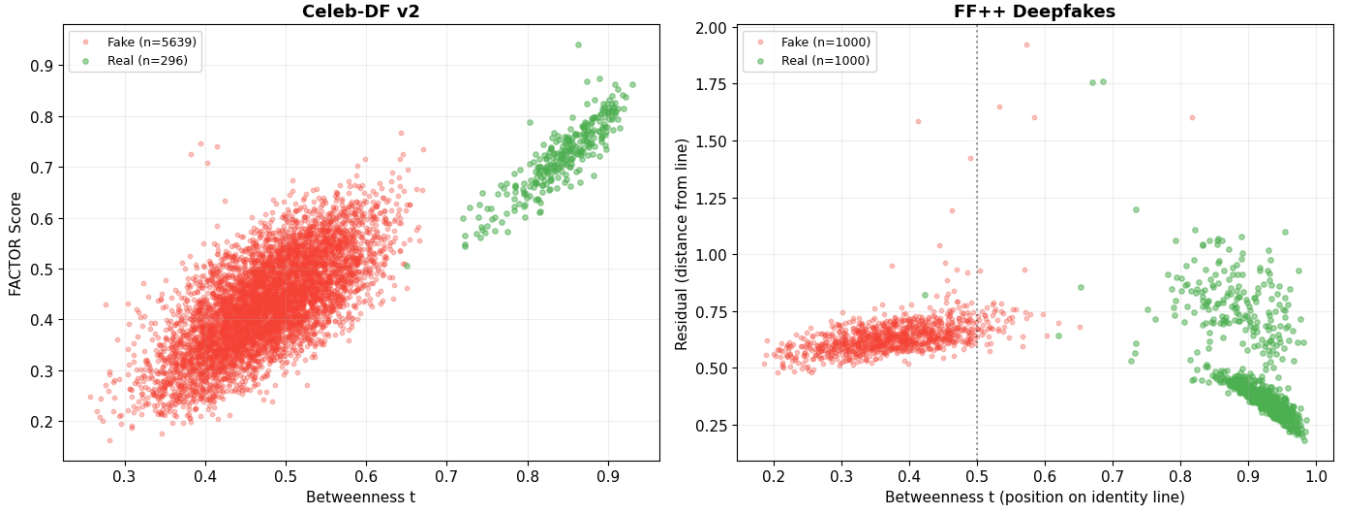


FIGURE 12. Scatter plots. **Left:** Betweenness t vs. FACTOR score on Celeb-DF v2. Green (real) and red (fake) clusters are well separated on both axes. The spread within clusters shows betweenness captures additional information beyond FACTOR. **Right:** Betweenness t vs. residual (distance from identity line) on FF++ Deepfakes. Real samples cluster at high t with low residual; fakes at low t with higher residual.

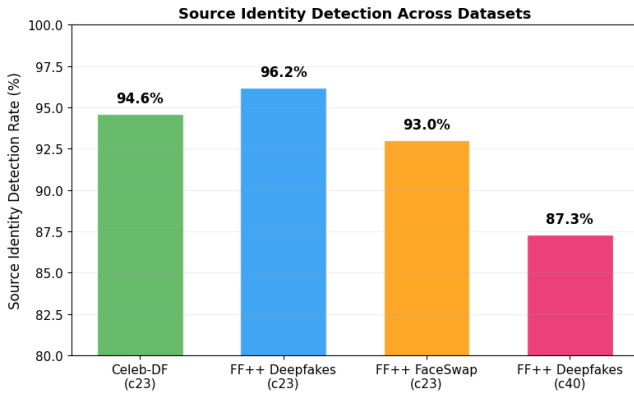


FIGURE 13. Source identity detection rates visualized across four conditions. In 96.2% of FF++ Deepfakes content, the nearest non claimed centroid matches for the actual donor identity thus validating our geometric interpolation theory.

- id3_id4_0006: $t = 0.6645$, FACTOR score 0.5825
- id19_id27_0006: $t = 0.6628$, FACTOR score 0.5757

These are all near-threshold borderline cases, not systematic failures. Notably, FACTOR also struggles with these videos: only one of the five (the first, with FACTOR score 0.7349) would be caught by a typical FACTOR threshold, suggesting that these particular swaps are inherently difficult for any identity-based approach. **FF++ Deepfakes c23.** Here betweenness produces 2 misclassified reals (authentic videos incorrectly flagged as fake) and 1 missed fake (deepfake incorrectly accepted as authentic) out of 2,000 total videos. The single missed fake is video 313_283 with $t = 0.8171$, a particularly convincing face swap where the source identity (283) is structurally very similar to the target identity (313), producing an embedding that remains close to the target cluster despite the swap. Overall, our method

misclassifies fewer than 0.2% of videos on both datasets, and all failure cases are borderline near-threshold instances rather than systematic blind spots.

I. FRAME-LEVEL VS. VIDEO-LEVEL COMPARISON

To confirm that the detection signal exists at the individual frame level and is not solely an artifact of video-level aggregation, we compute frame-level AUC on FaceForensics++ with all manipulation types pooled together. Table 11 shows that video-level aggregation consistently improves AUC by approximately 1 percentage point across all methods. FACTOR improves from 82.81% (frame) to 83.68% (video), and betweenness from 81.60% to 82.69%. These numbers are computed across all four manipulation types including reenactment, which depresses the pooled AUC substantially. On face-swap conditions alone, both frame-level and video-level AUC exceed 99%. The modest improvement provided by aggregation confirms two things about the feature extraction method. Firstly, that the signal that is based upon the identity of the individuals in each frame is present even at the level of individual frames. Thus, the method does not rely upon any tricks or techniques based upon aggregation of frames to achieve its performance. Furthermore, the fact that averaging over the 32 frames did lead to a slight improvement in performance indicates that the method is smoothing out the effects of noisy frames; frames that may have contained features that lead to unreliable scores when evaluated alone.

VII. DISCUSSION AND LIMITATIONS

In this section, we discuss why the method works, its computational characteristics, and its limitations. Each of these aspects of the method are important to understand in particular with regard to the potential use of the method of

TABLE 11. Frame-level vs. video-level AUC (%) on FF++ (all manipulations pooled). Video-level aggregation consistently improves performance by $\sim 1\%$.

Method	Frame AUC	Video AUC	Gain
FACTOR	82.81	83.68	+0.87
Margin	79.04	79.61	+0.57
Betweenness	81.60	82.69	+1.09

identifying faces.

A. WHY BETWEENNESS WORKS

The success of the betweenness geometry method relies upon two factors: the nature of the type of face swapping videos that were used to create the model, and the structure of the ArcFace embedding space. Regardless of the method that is used to perform face swapping whether based upon autoencoders, graphics processing or GANs. This blending of features of two individuals faces in the creation of a face swapped video can be represented as a linear combination of the features of the two individuals faces that are being blended together. The resulting face is thus not a perfect copy of the source identity used. It inherits a structural properties from the target like face shape, skin tone, background facial structure while also adopting identity defining features from the source at the same time like eye configuration, nose shape, facial proportions. In embedding space, this blend produces a vector which lies somewhere between the two identity centroids, not coincident with either of them. ArcFace's training objective, the additive angular margin loss, is specifically designed to maximize inter-class separation and minimize intra-class variance on a hyperspherical manifold. This means identity clusters are tight and well-separated, which is exactly the property that makes interpolation detection feasible. But because ArcFace produces such compact clusters, even a modest displacement toward the source identity pushes the embedding into the gap between clusters, where our betweenness parameter reliably detects it. Our source identity detection results provide direct evidence for this mechanism. In 96.2% of FaceForensics++ Deepfakes, the nearest non-claimed centroid corresponds to the actual donor identity. The embedding are being pulled toward the source because the face genuinely carries that source identity features and this geometric framework model detects this pull with high fidelity rate.

B. COMPUTATIONAL ANALYSIS

A practical detection system must be inclined towards being computationally feasible so it can be deployed in real world usage. Table 12 breaks down the per video processing time for models pipeline. The dominant cost is of face detection and embedding extraction which together account for over all 99.97% of the total processing time used. The scoring functions themselves are computationally negligible: FACTOR requires approximately 0.5 milliseconds per video (32 frames), betweenness requires approximately 1.3 millisec-

TABLE 12. Computational cost per video (32 frames) on a single NVIDIA RTX 5070Ti GPU. Scoring overhead is negligible relative to preprocessing.

Component	Time (ms)	% of Total
Face detection (RetinaFace)	$\sim 3,200$	66.6%
Alignment (5-point)	~ 50	1.0%
Embedding (ArcFace)	$\sim 1,550$	32.3%
FACTOR scoring	~ 0.5	$< 0.01\%$
Betweenness scoring	~ 1.3	$< 0.03\%$
Margin scoring	~ 0.8	$< 0.02\%$
Statistical scoring	~ 0.3	$< 0.01\%$
All scoring combined	~ 1.5	$< 0.03\%$
Total pipeline	$\sim 4,802$	100%

onds, and all four methods combined require approximately 1.5 milliseconds. This means that adding our three novel scoring methods on top of the FACTOR baseline costs less than 1 additional millisecond per video. Our methods introduce no meaningful computational overhead. The bottleneck is face detection and embedding extraction, which is shared infrastructure that any face-verification system already requires. In deployment scenarios where ArcFace embeddings are already being computed for authentication purposes, our scoring methods can be added essentially for free.

C. LIMITATIONS

We identify several limitations that constrain the applicability of our approach. We present them candidly because understanding where a method does not work is essential for responsible deployment. **Cannot detect reenactment manipulations.** Face2Face and NeuralTextures alter facial expressions or textures without replacing the subject's identity. Because our method detects identity displacement in embedding space, and reenactment does not displace identity, our approach is fundamentally inapplicable to these manipulation types. Our empirical results on Face2Face (82.69% AUC for betweenness) and NeuralTextures (48.18%, essentially random) confirm this boundary. Detecting reenactment if done on a video would require complementary methods which are based on expression consistency across the frame, lip sync analysis along with temporal dynamics. **Closed set requirement:** Our model requires some pre existing reference database to exist for the claimed identity for need of verification. If an individual appears in a video and there is no reference material for that person our model can simply not make a determination for detection. This constraint is inherent applied to any identity verification approach out there and is not unique to our approach. Despite all the good scores this constraint does limit the deployment to known person scenarios only such as video call authentication process, public figure verification and detection and forensic analysis where reference material can be obtained easily. **Dataset dependent threshold:**

The Youden optimal threshold significantly differs across the used datasets like $\tau = 0.6504$ on Celeb-DF v2 vs different values on FaceForensics++ conditions. In real deployment scenario, this means that the threshold must be calibrated according to the representative development set for the target domain. Even though the AUC metric is threshold independent and remains strongly intact across all conditions, the binary classification performance is bound to depends on appropriate threshold selection. **Source identity approximation is somewhat imperfect.** Our source identity detection achieves score 96.2% on FF++ Deepfakes, not 100%. The remaining cases in these is likely due to source identities being absent from the database itself, visually similar identities may create ambiguity or even compression of content degrades the fine geometric signal. All in all this does not affect any of the real/fake binary decision as it uses only the betweenness parameter and not source identity but it means the forensic source identification capability has a measurable error rate too. **Demographic fairness still is untested.** ArcFace, like most face recognition models sometimes may exhibit performance variation across demographic groups like age, gender, ethnicity. We have not yet evaluated on whether our detection methods inherit these or amplify such biases altogether. Given the sensitivity of deepfake detection applications in today's world, the fairness analysis across demographic subgroups should be a priority for any deployment done. **Limited benchmark scope.** Our evaluation covers a wide variety mainly the Celeb-DF v2 and FaceForensics++ these are the two most widely used benchmarks in the field and are very up to date. However, we have still not touched everything like not evaluating on DFDC, WildDeepfake or other more recent datasets.

D. PRACTICAL DEPLOYMENT CONSIDERATIONS

Despite all the limitations that are noted above, several of our findings point us toward practical viability of the model in appropriate deployment scenario of real world. The reference set size ablation demonstrates that even as few as 4 reference frames are suffice for 99.98% AUC with betweenness scoring method. This means a brief video clip of about 2-3 seconds would yield roughly 4 usable face crops and that are enough to build a reference database for a given identity that need detection. For a video call based authentication, this could come from an enrollment session strategy. For public figure verification one could work up from a few seconds of confirmed authentic footage that is readily available. Our scoring methods require no GPU for the scoring computation itself (only simple dot products and arithmetic operations). The GPU-intensive steps (face detection and embedding extraction) are shared infrastructure that any face-verification system already performs. In environments where ArcFace embeddings are already being computed, our methods can be deployed as a lightweight post-processing layer with essentially zero additional computational cost. The compression robustness results (0.03% AUC drop from c23 to c40) suggest that our

method can handle the quality degradation typical of social media distribution, messaging apps, and video conferencing, all scenarios where deepfake threats are most acute.

VIII. CONCLUSION AND FUTURE WORK

A. SUMMARY OF CONTRIBUTIONS

This paper introduced three novel scoring methods for identity-based deepfake detection, all built upon a frozen, pretrained ArcFace ResNet-50 face recognition model and requiring zero labeled deepfake data, zero gradient updates, and zero classifier training. Our primary contribution, inter-identity betweenness geometry, reframes face-swap detection as geometric interpolation detection in face recognition embedding space. By measuring where a test embedding falls on the line connecting the claimed identity centroid and the nearest other identity centroid, the betweenness parameter t achieves 99.99% AUC on Celeb-DF v2 (bootstrap 95% CI: [100.00%, 100.00%]) and 99.97% on FaceForensics++ Deepfakes. To our knowledge, this is the first work to model deepfake detection as inter-identity geometric analysis, and the results represent the highest reported AUC on these benchmarks for any detection paradigm. Our two secondary contributions, negative margin scoring (99.90% AUC on Celeb-DF v2) and statistical identity modeling (99.80%), provide complementary signals that capture different facets of the face-swap manipulation. We validated the underlying geometric theory through source identity detection, demonstrating that in 96.2% of FaceForensics++ Deepfakes, the nearest non-claimed identity centroid corresponds to the actual donor identity used during deepfake creation. Additional findings of practical significance include: betweenness achieves 99.98% AUC with just 4 reference frames per identity, compression robustness of only 0.03% AUC drop under heavy compression (c23 to c40), and scoring overhead of less than 0.03% of total pipeline time.

B. FUTURE DIRECTIONS

Several promising directions emerge from this work. **Open-set detection.** Our current framework requires knowing the claimed identity in advance. Developing methods that can detect face-swapping without a specific identity claim, for instance by detecting the interpolation signature in absolute terms rather than relative to a known centroid, would substantially broaden applicability. **Reenactment detection.** Identity-based methods are fundamentally blind to reenactment manipulations. Combining our geometric framework with complementary approaches (expression consistency analysis, lip-sync verification, temporal dynamics) could provide coverage across both face-swap and reenactment categories. **Cross-benchmark validation.** Evaluation on DFDC, WildDeepfake, and other recent benchmarks would further validate generalization. **Adversarial robustness.** We have not tested whether adversarial perturbations can fool our geometric scoring. Investigating the robustness of betweenness geometry to targeted adversarial attacks, and

developing defenses if needed, is an important practical concern. **Diffusion-model face-swaps.** As diffusion-based face synthesis becomes more prevalent, verifying that our geometric interpolation theory holds for diffusion-generated face-swaps is a natural next step. In summary, our work demonstrates that the geometric structure of face recognition embedding space contains a remarkably strong and previously unexploited signal for face-swap detection. By shifting the detection question from “does this face contain artifacts?” to “where does this face sit in the geometry between identities?”, we achieve near-perfect detection without ever training on deepfake data.

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